Tense and perspective: A solution to the ⌜*then*⌝-present puzzle*

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September 1, 2025

Abstract

In this paper, we focus on the cross-linguistic incompatibility between the temporal adverbial ⌜*then*⌝ and the present tense. We show that the pattern holds in both matrix and embedded environments, across an array of languages (including Modern Greek, Modern Hebrew, Russian and Japanese) each behaving differently with respect to tense shift. To account for this incompatibility, we propose that interpretation of tenses is sensitive to a perspective parameter. Tense shift occurs thanks to a perspective-shifting operator, similar to operator-based analyses of indexical shift (Anand and Nevins 2004; Anand 2006; Deal 2020). The main advantage of this account over alternative theories of tense shift, such as relative tense (Ogihara, 1996, 2022) or tense deletion (Ogihara and Sharvit, 2012), is that we are able to distinguish between shifted and deleted tense. This is needed to explain the incompatibility of ⌜*then*⌝ with the shifted present but not with deleted past. Finally, we argue that the newly introduced perspective parameter needs to be separate from the speech context and the evaluation index.

Keywords: Tense shift, Present, Deleted tense, Perspectives, Then, Temporal adverbials

1 Introduction

Then has many usages. It can be used anaphorically and deictically in discourse (Schiffrin (1992); Anand and Toosarvandani (2019) a.o.), in conditionals (Schiffrin, 1992; Iatridou, 1993), as well as as a temporal adverbial

*We thank Daniel Altshuler, Amir Anvari, Chris Barker, Amy Rose Deal, Philippe Schlenker, Yael Sharvit, Anna Szabolcsi, Gary Thoms, audiences at MACSIM, WCCFL, CLS, NYU, and MIT, three anonymous reviewers, as well as all of our consultants for their time and judgments. Both authors are first authors. Part of this work has been written by the same authors as a 10-page WCCFL proceedings paper available here: `https://ling.auf.net/lingbuzz/008398`.

(Kusumoto, 1999; Ogihara and Sharvit, 2012; Vostrikova, 2019; Tsilia, 2021; Tsilia and Zhao, 2025). As a discourse marker, its position is crucial, since clause-initial and clause-final *then* convey different readings. Schiffrin (1992) shows that clause-initial *then* shifts the reference time and tends to combine with perfective verbs, while clause-final *then* continues the reference time and combines with imperfective verbs.

(1) a. I had a fever and then I had a chill.
    b. I had a fever and I had a chill then.

Schiffrin 1992: (9)

In (1a) the chill occurred after the fever, while in (1b) *then* continues the reference time, with the chill occurring simultaneously with the fever. A similar distinction is observed in conditional *then* (Schiffrin, 1992). Similarly, Anand and Toosarvandani (2019) argue that both *now* and *then* have different meanings in clause-initial as opposed to clause-final positions.

(2) People began to leave. Then, the room was empty. Anand and Toosarvandani 2019: (36)

(3) The janitor turned off the lights. The room was empty then. Anand and Toosarvandani 2019: (40)

If we call the first sentence S' and the sentence with *then* S, with clause-initial *then*, the situation described by S temporally follows that by S', but with clause-final *then* they overlap. What is more, clause-initial *then*, contrary to clause-initial *now* can receive an interpretation where S precedes S'. This is the so-called backshifted interpretation:

(4) How things have changed since the campaign! Cohen plead guilty last week and incurred Trump's wrath. Then, he enjoyed his full support. Anand and Toosarvandani 2019: 37

Cohen's enjoying Trump's support can precede his incurring his wrath. Thus, S can precede S'. It is thus clear that *then* as a discourse marker has many interesting usages. Conditional *then* as well exhibits a clause-initial/clause-final distinction (Schiffrin, 1992), while also being compatible with the present (Iatridou, 1993):

(5) If Peter doesn't run for president, then I don't know what we are gonna do.

(adapted from Iatridou 1993)

However, *then* has one more usage as a *temporal* adjunct (i.e., *(during) then*), where it co-occurs with a matrix or an embedded tense, usually in clause-final position:

(6) 2 years ago, John knew that Mary was pregnant then.

It is this temporal usage of *then* that we will focus on in this paper; we put aside its function as a discourse marker and its usage in conditionals, as well as all other usages of *then*. We focus on the observation that *then* is incompatible with the present (Ogihara and Sharvit, 2012; Vostrikova, 2019; Tsilia, 2021; Tsilia and Zhao, 2025) and we show that it is compatible with a deleted past. Based on these data, we propose an account of tense shift which uses a perspective parameter to distinguish between shifted present and deleted past, while also capturing shift together effects in the temporal domain.

1.1 The puzzle

The temporal adverbial *then* cannot be combined with a *present*-tensed verb phrase, as shown by the contrast between (7) and (8).[1]

(7) Mary is feeling sick (*then).

(8) a. Mary was feeling sick (then).
b. Mary will be feeling sick (then).

At first glance, this contrast is not so hard to explain. The present in simple root clauses typically denotes a time overlapping with the time of utterance, while the past and the future denote times different from it. Assuming that the adverbial *then* has to refer to a time disjoint from the time of utterance, the unacceptability of (7) follows; indeed, we cannot use *then* to refer to a time overlapping with Mary's current illness. By contrast, the examples in (8) are acceptable because both the past and the future denote times that can be restricted by the temporal adverbial *then*.

Things become more interesting when we turn to embedded tense. In languages such as Modern Greek, Modern Hebrew, Japanese and Russian, an embedded present does not necessarily overlap with the time of utterance; it can be shifted. As shown in (9) for Greek, the present denotes a time overlapping with the past tense attitude. Yanis' knowledge co-occurs with Maria's pregnancy (in 2000). We will refer to this phenomenon as **tense shift**, and to a present tense with a shifted interpretation as a **shifted present**.

(9) To 2000, o Yanis **iksere** oti i Maria **ine** egkios.
The 2000, DET Yanis **know.PAST** that the Maria **be.PRES** pregnant
'In 2000, Yanis knew that Maria was (LIT. is) pregnant.' Modern Greek

Our previous explanation of the incompatibility between *then* and the present is too simplistic to explain the behavior of *then* in embedded clauses. If *then* is only required to denote a time disjoint from the time of *utterance*, then a shifted present should in principle be compatible with it. However, as first noticed by Ogihara and Sharvit (2012) for Modern Hebrew, Vostrikova (2019) for Russian and Sharvit (2018); Tsilia (2021) for Modern Greek, this is not the case:

(10) Lifney alpayim šana, Yosef **xašav** še Miriam **ohevet** oto (***az**).
Before 2,000 year, Yosef **think.PAST** that Miriam **love.PRES** him **then**
'2,000 years ago, Yosef thought that Miriam loved (LIT: loves) him then.'
Modern Hebrew (Ogihara and Sharvit 2012: fn. 6)

(11) V 2016 godu Tanja skazala, čto Putin (***togda**) ∅ prezident Rossii.
in 2016 year Tanja say.PAST that Putin **then** **be.PRES** president.NOM Russia.GEN
Intended: 'In 2016 Tanja said that Putin was (LIT. is) the president of Russia then.'
Russian (Vostrikova 2019: (4))

[1] This paper focuses on the *temporal* adverbial *then* and the ordinary present. We do not discuss the historical present, for which we refer the reader to Anand and Toosarvandani (2020, 2022).

(12) To 2000, o Yanis **iksere** oti i Maria **ine** egkios (***tote**).
The 2000, DET Yanis **know.PAST** that the Maria **be.PRES** pregnant **then**
Intended: 'In 2000, Yanis knew that Maria was (LIT. is) pregnant then.'
Modern Greek (Tsilia 2021: (78))

Despite the fact that in these languages the present is shiftable and does not necessarily denote the time of utterance, *then* is still incompatible with it. The incompatibility between *then* and the present in embedded environments is the core of **the ⌜*then*⌝-present puzzle**, as stated in (13).[2] The overarching goal of this paper is to use the ⌜*then*⌝-present puzzle as a window for a better understanding of tense semantics and tense shift.

(13) **The ⌜*then*⌝-present puzzle**
The temporal adverbial ⌜*then*⌝ cannot be adjoined to a present-tensed verb phrase, regardless of whether the present tense is shifted or not.

In previous literature the puzzle has mostly been discussed for present-under-past (Ogihara and Sharvit, 2012; Vostrikova, 2019), while here we further consider present-under-future, and more importantly we observe that ⌜*then*⌝ is incompatible with a shifted present, but compatible with a deleted past. This teaches us that, contrary to what is commonly assumed, the shifted present is not semantically equivalent to the deleted past. Our proposal will capture their difference.

1.2 Overview

The semantic contribution of a tense morpheme is to establish a **temporal reference**, which is in turn related to a **temporal anchor**, the time of the denoted event/state.[3] A present tense refers to a time *overlapping* with the temporal anchor, while a past tense refers to a time *preceding* it. As for tense shift, there are two main ways to account for it. The first one, exemplified by Ogihara (1996, 2022), argues that tense shift is the result of the temporal anchor shifting. In that view, an embedded shifted present is thus anchored to a time distinct from the time of utterance; it is anchored to the time of the matrix attitude, as in (9). Theories implementing this idea treat the temporal anchor as a (covert) pronoun, analogous to the temporal reference. Every tense morpheme therefore carries two indices, one for its reference and one for its anchor. Accordingly, the sentence in (9), repeated with English pseudo-glossing in (14), corresponds to the following Logical Form (henceforth LF) where the anchor of the embedded, shifted present is bound by the reference of the matrix past.

(14) Yanis knew that Maria is pregnant. (Lit. from Modern Greek)
Yanis know-$\textbf{PAST}_{\underset{\text{REF}}{1},\ \overset{\text{ANCHOR}}{0}}$ [… Maria be-$\textbf{PRES}_{\underset{\text{REF}}{2},\ \overset{\text{ANCHOR}}{1}}$ pregnant] (to be rejected)

[2]Throughout this paper, we will use the corner quoted ⌜***then***⌝ as a placeholder for temporal adverbials across languages corresponding to English *then*. It will refer to adverbials as opposed to periphrastic anaphoric expressions (e.g. in Modern Greek, ⌜*then*⌝ corresponds to *tote* 'then', as in (12), as opposed to *ekino ton kero* 'at that time').

[3]We follow Partee (1973) in treating tenses as pronouns, as opposed to treating them as operators that quantify over domains of times (akin to the tense logic of Prior 1957, 1967). This choice will not be crucial for our main arguments.

The second approach, as discussed in, e.g. Ogihara and Sharvit (2012), resorts to a '**deletion**' mechanism for the shifted present. In this approach, the shifted present no longer has a temporal anchor, being bound by the reference of a higher tense. This corresponds to the LF in (15) for (9). This approach essentially identifies the shifted present with **deleted tenses**. Observed in languages such as English and Modern Greek, deleted tenses (such as the deleted past) give rise to the same kind of simultaneous reading as the shifted present. The past tense feature remains uninterpreted, or gets 'deleted' as in (16).

(15) Yanis knew that Maria is pregnant. (Lit. from Modern Greek)
Yanis know-$\textbf{PAST}_{1,0}$ [… Maria be-$\cancel{\textbf{PRES}}_1$ pregnant] (to be rejected)

(16) John found out that Mary **was** drunk.
John find-out-$\textbf{PAST}_{1,0}$ [… Maria be-$\cancel{\textbf{PAST}}_1$ drunk]
$\rightsquigarrow$ Mary was drunk at the time when John found out.

In this paper, we reject both of these options, and adopt the LF in (17). The temporal anchor is no longer modeled as a pronoun, but as an interpretation **parameter** π which we call the temporal **perspective**. Therefore, tenses now only bear a single index corresponding to their temporal reference; tense features restrict the range of this temporal reference with respect to π. Tense shift is achieved thanks to a perspective-shifting operator $\textbf{OP}_\pi$, which takes scope over the entire embedded clause, and works by changing the value of π used to interpret the embedded clause.

(17) Yanis knew that Maria is pregnant. (Lit. from Modern Greek)
$[\![\text{Yanis know-}\textbf{PAST}_{\underset{\text{REF}}{1}}\ [\boxed{\textbf{OP}_\pi}\ \text{Maria be-}\textbf{PRES}_{\underset{\text{REF}}{2}}\ \text{pregnant}]]\!]^{\cdots\ \overset{\text{ANCHOR}}{\pi}\ \cdots}$

The interaction between tenses and ⌜*then*⌝ will lead us to prefer LFs like (17) over the first type of approach, e.g., (14), as well as over the second type of approach, e.g., (15) and (16). (17) is similar to (14) in attributing tense shift to the shift of the anchor. However, treating the anchor as an interpretation parameter as in (17) has an advantage, granting it a 'global' effect on all expressions within the domain of interpretation. We will show that expressions within the same minimal domain **shift together**, and can only be shifted by a propositional operator like $\textbf{OP}_\pi$. Making the perspective a parameter allows (17) to generate only a subset of readings that (14) does, a crucial step towards solving the ⌜*then*⌝-present puzzle. We will expand on this in Section 2.3. Readers familiar with the literature on **indexical shift** may also recognize the parallel between (17) and the theory developed in Anand and Nevins (2004) and Anand (2006). Indeed, as will become clear in Section 4, our theory draws heavily on their analysis of the **Shift Together** effect of indexicals.

As for the deletion approach (see (15), (16)), it identifies the shifted present with deleted tense, wrongly predicting that ⌜*then*⌝ should behave the same with both type of tenses. In Section 3 we will show that this is empirically wrong; in fact, ⌜*then*⌝ is compatible with deleted, but not with shifted tense in Modern Greek. Thus, a distinction needs to be made between deleted and shifted tense. We propose that only shifted tense is sensitive to the perspective. To our knowledge this is the first empirical evidence demonstrating that shifted tense differs from deleted tense. In Section 6, we propose an analysis that captures this difference in terms of perspectival sensitivity.

Finally, we will argue that the temporal perspective parameter π is different from other interpretation parameters, such as the **context** of utterance, and the **index** used to evaluate the truth-conditional content (e.g. worlds, times). We will show that our data cannot be fully accounted for without a separate perspective parameter. We will only delve into the temporal component of the perspective parameter, although there is definitely more to explore in terms of other 'perspective-sensitive items', see e.g. Partee (2004), Oshima (2005), McCready (2007), Pearson (2013), Bylinina (2014), Sudo (2015), Sundaresan (2018) and many others. In fact, our formalism closely resembles the one in Sudo's (2015) analysis of locative perspectives and the semantics of motion verbs. Exploring connections between temporal and other kinds of perspective-sensitive expressions is an interesting direction for future research.

The rest of the paper is structured as follows. In Section 2, we present the empirical picture of the ⌜*then*⌝-present puzzle. We show that the puzzle arises both with shifted and non-shifted present, across a number of languages that constitute a complex typology of tense shift. We then argue that the temporal anchor should be an interpretation parameter. Section 3 shows that deleted tense and shifted present cannot be semantically equivalent. This difference is theoretically accounted for, while also solving the ⌜*then*⌝-present puzzle in Section 4. We present a compositional fragment, building on the theory of indexical shift. We then apply this to our data in Section 5. The account also proposes a new way to treat tense shift cross-linguistically. In Section 6 we delve into the difference between deleted and shifted tense in our system, arguing that only shifted tenses are sensitive to the perspective. Section 7 establishes why the perspective parameter is necessary, and crucially distinct from the context of utterance and the evaluation index. Section 8 concludes.

2 The shifted present and the ⌜*then*⌝-present puzzle

The languages surveyed in this paper include English, Modern Greek, Modern Hebrew, Japanese and Russian. These are all languages with tense, as well as a representative sample of the typology of embedded tense (cf. Ogihara and Sharvit 2012). In this Section we focus on the embedded present. Depending on the tense of the main clause, and on the nature of the embedded clause (i.e. whether it is a complement of a speech/attitude verb or a relative clause), languages may or may not shift the embedded present. We go through each configuration, starting with the present embedded under *past*-tensed matrix verbs and moving on to the present embedded under *future*. Regardless of whether the present can be shifted, it is always incompatible with the adverbial ⌜*then*⌝. Based on this observation, we argue that the temporal anchor should be treated as a parameter as in (17) rather than as a pronoun as in (14).

2.1 Present-under-past

Under a past-tensed matrix verb, the embedded present in English is always indexical, i.e., it cannot shift, with the event in the embedded clause being interpreted at the time of utterance:

(18) In his childhood, Joseph **met** a woman who **loves** traveling
⇝ the woman loves traveling *at the time of utterance* (Ogihara and Sharvit 2012: (19a))

(19) A month ago, John **found** out that Mary **loves** him.
⇝ Mary loves John *at the time of utterance* (Ogihara and Sharvit 2012: (15a))

In a relative clause, there is no *simultaneous* reading between the matrix and the embedded verb. In (18), for example, the woman that loves traveling *now* may very well not have loved it when she met Joseph. When we move to verb complements, there is the famous 'double-access' reading (Smith, 1978). In (19), for example, Mary loved John a month ago but still loves him at the time of the utterance. The English present is still read indexically, and the 'double-access' reading follows from other constraints.[4] If we pragmatically rule out the double-access reading, a pure simultaneous reading of the embedded present is unavailable:

(20) #Two thousand years ago, Joseph **found** out that Mary **loves** him.
(Ogihara and Sharvit 2012: (14))

Therefore, the English present is unshiftable, receiving an indexical reading in both root and embedded clauses. It should thus not come as a surprise that it is incompatible with *then* in embedded environments, just like in unembedded ones (see (7)):

(21) *A month ago, John **found** out that Mary **loves** him **then**.

(22) *In his childhood, Joseph **met** a woman who **loves** traveling **then**.

Unlike English, however, languages such as Modern Greek, Modern Hebrew and Russian have a shiftable present. Recall from (12), repeated below as (23) with the addition of (24) and (25), that a present embedded under a **past tense speech/attitude verb** can be shifted, giving rise to a simultaneous reading. Contrary to English (see (19) and (20)), a shifted present does not force the event to hold at the time of utterance. Despite this, it is still incompatible with ⌜*then*⌝.

[4]For instance, Abusch (1988, 1997) proposed a pragmatic 'Upper Limit Constraint' that prohibits embedded tenses from being interpreted relative to any time later than the attitude holder's 'now'; it is through this constraint that the double-access reading is obtained, although a full discussion of this lies outside the scope of the paper. For alternative approaches, see e.g. Ogihara (1995, 1996); Altshuler and Schwarzschild (2013); Bary and Altshuler (2015); Klecha (2018). Most of these approaches subscribe to the view that the English present embedded under past is anchored to the time of utterance, with Altshuler and Schwarzschild (2013) arguing that the English present is both deictic and relative. See also Kusily (2020, 2024) who shows that simultaneous readings can marginally be obtained in English VP-fronting constructions:

(1) Think that Mary is angry, John did. (Kusily 2020: (8))

We put these constructions aside for the purposes of this paper, treating the English present as always referring to the time of the utterance. That being said, it would be interesting to see if the puzzle holds in English VP-fronting constructions.

(23) To 2000, o Yanis **iksere** oti i Maria **ine** egkios (***tote**).
The 2000, DET Yanis **know.PAST** that the Maria **be.PRES** pregnant **then**
'In 2000, Yanis knew that Maria was (LIT. is) pregnant then.'

Modern Greek (Tsilia 2021: (78))

(24) Lifney alpayim šana, Yosef **xašav** še Miriam **ohevet** oto (***az**).
Before 2,000 year, Yosef **think.PAST** that Miriam **love.PRES** him **then**
'2,000 years ago, Yosef thought that Miriam loved (LIT: loves) him then.'

Modern Hebrew (Ogihara and Sharvit 2012: fn. 6)[5]

(25) V dvuxtysjačnom godu Ivan **znal**, čto (***togda**) Maša ∅ beremenna.
In 2000 year Ivan **know.PAST** that **then** Maša **be.PRES** pregnant
'In 2000, Ivan knew that Masha was (LIT: is) pregnant then.'

Russian (Tsilia 2021: (71), replicating Vostrikova 2019)

On the other hand, Modern Greek, Modern Hebrew and Russian are like English in *not* allowing tense shift in **relative clauses**. The embedded present in (26)–(28) unambiguously refers to a time that overlaps with the time of utterance like in (18); and like (22), it is incompatible with ⌜*then*⌝.

(26) Prin 20 chronia o Pavlos **sinerghastike** me enan andra pu **ine** proedros (***tote**).
Before 20 years the Pavlos **collaborate.PFV.PAST** with a man who **be.PRES** president **then**.
Only: '20 years ago, Pavlos collaborated with a man who is president (now).'

Modern Greek (adapted from Tsilia 2021: (32))

(27) be-yalduto, **pagaš** Yosef iša še **ohevet** letayel (***az**).
in-childhood-his **meet.PAST** Yosef woman COMP **love.PRES** traveling **then**
Only: 'In his childhood, Yosef met a woman who loves traveling (now).'

Modern Hebrew (adapted from Ogihara and Sharvit 2012: (20a))

(28) Maša **videla** čeloveka, kotoryi (***togda**) **plačet**.
Maša **see.PAST.IMP** man who **then** **cry.PRES**
Only: 'Maša saw a man who is crying (now).'

Russian (adapted from Kusumoto 1999: (129a))

Finally, Japanese differs from all the languages above in allowing the present to shift not only in the complement of speech/attitude verbs as in (29), but also in relative clauses as in (30) (see also Ogihara, 1996; Ogihara and Sharvit, 2012). In both environments the ⌜*then*⌝-present puzzle arises.

(29) ninenn mae, Yusuke-wa Yukiko-ga (***tooji**) ninnshinn **shite-iru** to **shitteita**.
2-years before Yusuke-TOP Yukiko-NOM **then**) pregnant **be-PRES** with **know.PAST**.
'2 years ago, Yusuke knew that Yukiko was (LIT: is) pregnant.' Japanese

[5] Ogihara and Sharvit (2012) noted that the ⌜*then*⌝-present configuration is unacceptable to '*many*, but admittedly not all' native speakers. We will not attempt to capture this variation.

(30) ninenn mae, Yusuke-wa (***tooji**) seifu de hatarai-te-**iru** hito to renkei o
2-year before Yusuke-TOP **then** government LOC work.PROG.**PRES** person with collaboration OBJ
hakat-te-**ita**.
plan.IMPFV.**PAST**.
'Two years ago, Yusuke collaborated with a man who was (LIT: is) president.' Japanese

The languages under consideration can therefore be divided into three subgroups: (i) English, where the present does not shift under past, whether in the complement of a speech/attitude verb or in relative clauses, (ii) Modern Greek, Modern Hebrew and Russian, where the present may shift in speech/attitude reports but not in relative clauses, and (iii) Japanese, where the present may shift in both environments. The ⌜*then*⌝-present puzzle arises consistently in all embedded constructions; wherever a language allows for a shifted present, the puzzle arises. The findings are summarized below in Table 1.

LANGUAGE	ENVIRONMENT	shifted present?	⌜*then*⌝-present
Modern Greek	attitude report	✓	incompatible
	relative clause	✗	incompatible
Modern Hebrew	attitude report	✓	incompatible
	relative clause	✗	incompatible
Russian	attitude report	✓	incompatible
	relative clause	✗	incompatible
Japanese	attitude report	✓	incompatible
	relative clause	✓	incompatible
English	attitude report	✗	incompatible
	relative clause	✗	incompatible

Table 1: PRESENT UNDER PAST

2.2 Present under future

When the matrix verb is in the future tense, languages behave more uniformly. Almost all surveyed languages allow for a shifted present under future, both in speech/attitude reports and in relative clauses. With the exception of the English present, the puzzle arises with all other presents shifted under future we surveyed.

(31) Present under future: attitude reports

a. Modern Greek

To 2030, i Maria **tha** pi oti **ine** sti filaki (***tote**).
In 2030, the Maria **FUT** say that **be.PRES** to-the jail **then**
'In 2030, Maria will say that she is in jail (in 2030).' (Tsilia 2021: (79))

b. Modern Hebrew

Be 2030, John **yagid** še hu ∅ ba kele (***az**)
In 2030, John **say.FUT** that he **be.PRES** in jail **then**

'In 2030, John will say that he is in jail (in 2030).' (Tsilia 2021: (76))

c. Russian

V 2030, Ivan **skažet**, čto on (***togda**) **sidit** v tjur'me.
In 2030, Ivan **say.FUT**, that he **then** **sit.IMPFV.PRES** in jail
'In 2030, Ivan will say that he is in jail (in 2030).' (Tsilia 2021: (72))

d. English

In 2030, John will say that he is in jail (%**then**). (Tsilia 2021: (85))

(32) Present under future: relative clauses

a. Modern Greek

To 2030, i Zoi **tha** ine pantremeni me kapion pu **ine** (***tote**) sti filaki.
In 2030, the Zoi **FUT** be married to someone who **be.PRES** **then** to-the jail
'In 2030, Zoe will be married to someone who is in jail.' (Tsilia 2021: (80))

b. Modern Hebrew

Be 2030, John **yihié** nasui le miši še ∅ ba kele (***az**).
In 2030, John **be.FUT** married to someone.FEM that **be.PRES** in jail **then**
'In 2030, John will be married to someone who is in jail.' (Tsilia 2021: (77))

c. Russian

V 2030 godu, Ivan budet ženat na tom, kto (***togda**) **sidit** v tjur'me.
In 2030 year, Ivan **be.FUT** marry.PFV.PASS to someone, who **then** **sit.IMPFV.PRES** in jail
'In 2030, Ivan will be married to someone who is in jail.'

d. English

In 2030, John will be married to someone who is in jail (**then**).
(Tsilia 2021: (87), adapted from von Fintel and Heim (1997-2021): §7, (4))

The patterns are summarized in Table 2. Two comments are in order. First, *tooji* 'then' in Japanese is only used with past expressions, so it is ruled out even with a matrix future tense as shown in (33). This is why we did not include Japanese examples in (31) and (32).

(33) *Two friends are making plans to go to the movies on Saturday. One of them says before they leave:*

Sonotoki/*Tooji ai-mashou.
Then see-let's
'See you then.'

Secondly, notice that Modern Greek, Modern Hebrew, and Russian, all languages with a shiftable present exhibit the puzzle under future. However, in English, a language where the present can only shift under future, *then* is compatible with the present under future both in attitude reports as in (31d) (at least for some speakers), as well as in relative clauses as in (32d). We will come back to this issue in Sections 5.3 and 6.

We summarize the cross-linguistic patterns under future in attitude reports and in relative clauses in the table below:

LANGUAGE	ENVIRONMENT	shifted present?	⌜*then*⌝-present
Modern Greek	attitude report	✓	incompatible
	relative clause	✓	incompatible
Modern Hebrew	attitude report	✓	incompatible
	relative clause	✓	incompatible
Russian	attitude report	✓	incompatible
	relative clause	✓	incompatible
Japanese	attitude report	✓	N/A
	relative clause	✓	N/A
English	attitude report	✓	variation (%)
	relative clause	✓	compatible

Table 2: PRESENT UNDER FUTURE

2.3 An interpretation *parameter* for tenses and ⌜*then*⌝

Let us now return to the theoretical debate of whether the temporal anchor of tenses and ⌜*then*⌝ should be treated as an interpretation parameter instead of as a temporal pronoun. The relevant LFs, (14) and (17), are repeated below as (34) and (35). We have also included the assignment function g that assigns values to variables. Note that we use $\circ$ to indicate overlapping temporal intervals.

(34) Yanis knew that Maria is pregnant (Lit. from Modern Greek)
$[\![\text{Yanis know-}\textbf{PAST}_{\underset{\text{REF}}{1},\overset{\text{ANCHOR}}{0}}\ [\ldots\ \text{Maria be-}\textbf{PRES}_{\underset{\text{REF}}{2},\overset{\text{ANCHOR}}{1}}\ \text{pregnant}]]\!]^{\cdots g \cdots}$ (to be rejected)
$\rightsquigarrow g(2) \circ g(1)$

(35) Yanis knew that Maria is pregnant (Lit. from Modern Greek)
$[\![\text{Yanis know-}\textbf{PAST}_{\underset{\text{REF}}{1}}\ [\boxed{\textbf{OP}_\pi}\ \text{Maria be-}\textbf{PRES}_{\underset{\text{REF}}{2}}\ \text{pregnant}]]\!]^{\cdots \overset{\text{ANCHOR}}{\pi}, g \cdots}$
$\rightsquigarrow g(2) \circ (\pi = g(1))$

The main difference between the two LFs above is their treatment of the temporal anchor. In (34), the anchor is a (covert) temporal pronoun, represented as the second index on the tense morpheme. The shifted reading of the present is derived through pronominal binding between the reference of the matrix past $g(1)$ and the anchor of the shifted present. In (35), on the other hand, the temporal anchor is a perspective parameter π, and tense shift is the result of a propositional operator $\textbf{OP}_\pi$ which shifts the value of π to $g(1)$. We will set aside for now how $\textbf{OP}_\pi$ works (details to come in Section 4). Both LFs derive the shifted reading of the present, where the reference of the shifted present $g(2)$ overlaps ('$\circ$') with the temporal reference of the matrix past $g(1)$ (i.e. the time of Yanis' knowledge).

We now provide empirical support for the parametric treatment of the temporal anchor as in (35). Let us consider how the incompatibility between the shifted present and ⌜*then*⌝ could be accounted for in either approach. We assume that ⌜*then*⌝ is similar to a tense in being sensitive to the temporal anchor. Contrary to the present, it requires its reference to be *disjoint* from the temporal anchor. The LFs of the ungrammatical ⌜*then*⌝-present sentences for each approach are in (36) below:

(36) *Yanis knew that Maria is pregnant then.

a. $[\![$Yanis know-$\textbf{PAST}_{1,0}$ [… Maria be-$\textbf{PRES}_{2,1}$ pregnant $\textbf{then}_{3,1}$]$]\!]^{\cdots g \cdots}$ (pronominal anchor)

b. $[\![$Yanis know-$\textbf{PAST}_{\underset{\text{REF}}{1}}$ [$\boxed{\textbf{OP}_\pi}$ Maria be-$\textbf{PRES}_{2}$ pregnant $\textbf{then}_{3}$]$]\!]^{\cdots \pi, g \cdots}$ (parametric anchor)

In both LFs, the embedded present and ⌜*then*⌝ share the same temporal anchor valued at $g(1)$ (in the parametric approach the anchor of the present is equal to the perspective, i.e., $\pi = g(1)$). While the present requires its reference to overlap with the temporal anchor (i.e., $g(2) \circ (\pi = g(1))$), ⌜*then*⌝ requires its reference to be disjoint from it (i.e., $\neg(g(3) \circ (\pi = g(1)))$). As a result, because of these conflicting conditions, ⌜*then*⌝ cannot be used to restrict the embedded present (i.e. $g(2) \subseteq g(3)$ cannot hold), and so the puzzle is accounted for.

The advantage of (36b) is that it *guarantees* that the present and ⌜*then*⌝ share the same anchor, since $\textbf{OP}_\pi$ takes scope over the whole embedded clause. On the contrary, in (36a) it has to be *stipulated* that the present and ⌜*then*⌝ share the same anchor. Thus, there is an over-generation of unattested LFs with the pronominal approach of the temporal anchor. For example, in addition to (36a), (37) is also allowed:

(37) Yanis know-$\textbf{PAST}_{1,0}$ [… Maria be-$\textbf{PRES}_{2,1}$ pregnant $\textbf{then}_{3,0}$]

In this case, the shifted $\textbf{PRES}_{2,1}$ is anchored to the reference of the matrix past denoted by $g(1)$. Given that the matrix tense is a *past* tense, $g(1)$ has to *precede* its anchor, namely $g(0)$. Contrary to (36a), in the LF in (37) ⌜*then*⌝ shares its anchor $g(0)$ with the matrix past instead of the embedded present. This LF is unattested, since it would make ⌜*then*⌝ compatible with the embedded present. In (37), the reference of ⌜*then*⌝ $g(3)$ could be a temporal interval that contains the reference of the shifted present, $g(2)$. This is because $g(2)$ overlaps with $g(1)$, but $g(1)$ is disjoint from ⌜*then*⌝'s anchor $g(0)$. In other words, because in a pronominal system we are free to choose different anchors for ⌜*then*⌝ and the present, there is an LF which makes them compatible. In such a system, tense shift can *feed* the co-occurrence of the shifted present with ⌜*then*⌝. This over-generates, and fails to account for the puzzle. As we already saw, this is avoided in the parametric approach, since the whole embedded clause has to be interpreted with respect to the same anchor. We thus conclude that it is a better approach on the grounds that it accounts for the ⌜*then*⌝-present puzzle, and does not over-generate.

At this point, one could argue that ⌜*then*⌝ is not sensitive to a temporal anchor to begin with. If this were the case, we would have an LF (within the pronominal approach) where ⌜*then*⌝ is single-indexed, contrary to tenses which are perspective-sensitive:

(38) $[\![$Yanis know-$\textbf{PAST}_{1,0}$ [… Maria be-$\textbf{PRES}_{2,1}$ pregnant $\textbf{then}_{3}$]$]\!]^{\cdots g \cdots}$

How would we then account for the puzzle? If ⌜*then*⌝ is not anchored, there has to be some restriction on its reference $g(3)$, prohibiting it from combining with the shifted $\textbf{PRES}_{2,1}$. Apart from the fact that such a restriction would be *ad hoc*, it would also mean that anything with the reference $g(2)$ cannot combine with *then*$_3$ either. This is not the case. Many Modern Hebrew and Russian speakers accept the co-occurrence of ⌜*then*⌝ with an embedded *past*, which generates the same simultaneous reading (therefore the same temporal reference) as the shifted present:

(39) lifney alpayim šana, Yosef **xašav** še Miriam **ahava** oto **az**.
before two-thousand year Yosef **think**.**PAST** that Miriam **love**.**PAST** him **then**
'Two thousand years ago Yosef believed that Miriam loved him then.'
Modern Hebrew (Ogihara and Sharvit 2012: (7a))

(40) V 2016 godu Tanja **skazala**, čto **togda** Putin **byl** prezidentom Rossii.
in 2016 year Tanja **say**.**PAST** that **then** Putin **be**.**PAST** president.INST Russia.GEN.
'In 2016 Tanja said that Putin was the president of Russia then.' Russian (Vostrikova 2019: (2))

It is thus possible for ⌜*then*⌝ to have the same reference as the embedded tense (in (38) it is possible for $g(2) = g(3)$), even though it is not possible for ⌜*then*⌝ to co-occur with the present. But if, as we said earlier, a constraint banning $g(2) = g(3)$ is required to explain the puzzle, this constraint would also predict (39) and (40) to be ungrammatical, contrary to fact. This shows that an anchor-free treatment of ⌜*then*⌝ will not save the pronominal approach either.

A proponent of the pronominal approach could, as a last resort, come up with syntactic constraints forcing the anchor of the embedded tense to be equal to the anchor of ⌜*then*⌝ (e.g. the *local binding* constraint as in Anand 2007). Such a syntactic restriction merits further empirical investigation, but we point out two practical concerns. First, to implement this idea, one would need to establish a syntactic relation between two *covert* temporal pronouns, which seems intangible. Second, even if such a syntactic relation could be established, we would expect it to apply to all occurrences of a single tense. However, as we can see in (41), multiple occurrences of the same tense morpheme may correspond to different temporal references.

(41) a. Yesterday, John **found out** that Mary **was** drunk the day before.
$\rightsquigarrow$ John found out yesterday that Mary was drunk two days ago
(cf. the simultaneous reading in (16))

b. Yesterday, John **found out** that Mary **was** drunk the day before and **was** feeling sick.
$\rightsquigarrow$ John found out yesterday that Erica was drunk two days ago and was feeling sick yesterday.

In sum, to account for the ⌜*then*⌝-present puzzle, clausemate tenses and ⌜*then*⌝ should share the same temporal anchor, so that, when they shift, they shift together. This favors a parametric approach, which explains this shift together behavior very naturally. We will implement such a parametric approach by treating the temporal anchor as the perspective parameter. We will show how this results to a theory of tense that covers our data, including the ⌜*then*⌝-present puzzle. We will delve deeper into the nature of the perspective, how the operator $\textsc{op}_\pi$ works and how they interact with embedding environments in Sections 4–7. There, the solution to the ⌜*then*⌝-present puzzle will be presented along with an account of the tense-shift typology.

3 Deleted tense and ⌜*then*⌝

As mentioned in the Introduction, in addition to the pronominal approach, there is also a deletion approach to tense shift. This account invokes a deletion mechanism reducing the shifted present to a bound temporal pronoun, as sketched in (42).

(42) Yanis knew that Maria is pregnant. (Lit. from Modern Greek)
$[\![$Yanis know-$\textsc{past}_1$ [… Maria be-$\cancel{\textsc{pres}}_1$ pregnant]$]\!]^{\cdots\,\pi, g \cdots}$ (to be rejected)

In the LF above, we have also included the temporal anchor as π to show how the perspective would be implemented in such an approach. The strike-through on $\cancel{\textsc{pres}}_1$ deletes any kind of perspective-sensitivity, leaving the embedded present simply bound by the reference of the matrix past $g(1)$.

This deletion-based account, akin to what has been proposed in e.g. Ogihara and Sharvit (2012), essentially identifies the shifted present with **deleted past**. The latter also generates a simultaneous reading, through past-under-past constructions such as (43). The deletion mechanism is implemented with a sequence-of-tense (SoT) rule, formulated informally in (44).[6]

(43) John **found out** that Mary **was** drunk.
$[\![$John find-out-$\textsc{past}_1$ [… Maria be-$\cancel{\textsc{past}}_1$ drunk]$]\!]^{\cdots\,\pi, g \cdots}$
$\rightsquigarrow$ Mary was drunk at the time when John found out.

(44) Sequence-of-tense (SoT)
The tense feature of a tense c-commanded by an agreeing tense may be deleted at the Logical Form.

But if the SoT rule licenses tense deletion only on the lower of two *agreeing* tenses, how can the deletion of the present feature, as in (42), be licensed under a past tense? In addition to this concern, we show below that identifying the shifted present with the deleted past is empirically inaccurate, since the ⌜*then*⌝-present puzzle arises only with the former. At this point one may ask: how do we know deleted tense even exists? This is a valid concern, since 'simple' past-under-past constructions like (43) can have a *de re* LF which also gives rise to the simultaneous reading without tense deletion:

(45) John **found out** that Mary **was** drunk.
$[\![$John find-out-$\textsc{past}_1$ [Maria be-$\textsc{past}_2$ drunk]$]\!]^{\cdots\,\pi, g \cdots}$

[6] An alternative to deletion is that the lower tense originates as a 'zero' tense that lacks any tense feature, and the past tense is only morphological, being the result of agreement. For an extensive discussion on this option, see e.g. Kratzer (1998). Our arguments will not be affected by the choice between zero and deleted tense.

In (45) the embedded past is not deleted; there is also no tense shift either (since there is no $\textsc{op}_\pi$). Given a parameter π, at the absence of $\textsc{op}_\pi$, the matrix and the embedded past have to share the same temporal anchor, i.e. both $g(1)$ and $g(2)$ precede π. This leaves open the possibility of co-reference between the two tenses, i.e. $g(1) = g(2)$, which gives rise to the simultaneous reading. This is what Abusch (1997) called the 'temporal *de re*' reading. In order to control for this possibility, we will follow Abusch in considering the more complicated example in (46):[7]

(46) A week ago, John said that in ten days he would say to his girlfriend that they **were** meeting for the last time.

The point of this example is that the most deeply embedded past tense is not past with respect to any moment in the sentence, and therefore *has to* be deleted. (46) has the reading that John said a week ago that three days from now, he will say to his girlfriend: "we *are* now meeting for the last time". His utterance will be in the present tense, since it will coincide with the time of the meeting. But this has not happened yet; it will happen in three days. The most deeply embedded past therefore refers to a time that does not precede the time of utterance, or any other salient temporal reference established in the discourse. Such examples cannot be accounted for via temporal *de re*, since the past is not past with respect to the time of the utterance or the attitude. Thus, a deletion mechanism is necessary to account for such examples. The same phenomenon is observed in Modern Greek (Sharvit 2018; Tsilia 2021):

(47) Prin mia evdhomadha, o Jorghos ipe oti se dheka meres tha eleghe stin
Before one week the Jorghos say.PAST that in ten days will say.IMPFV.PAST to-the
kopela tu oti **sinadjiondusan** ja teleftea fora.
girlfriend of-his that meet.IMPFV.**PAST** for last time
'A week ago, Jorghos said that in ten days he would say to his girlfriend that they were meeting for the last time' Modern Greek (Sharvit 2018: (40))

Apart from English and Modern Greek, none of the other languages we consider (Modern Hebrew, Japanese and Russian) has deleted tense; the counterpart of (46) in these languages lacks the simultaneous reading.

Still, this could suggest that ⌜*then*⌝ is incompatible with the shifted present but is compatible with a *de re* past. This is essentially what Vostrikova (2019) proposes for Russian. However, such a *de re* analysis will not work for Modern Greek or English, since these are languages with a deleted past, and yet the deleted past is still compatible with ⌜*then*⌝. The novel puzzle here is the following: if the deleted past is semantically equivalent to the shifted present, as commonly assumed (e.g. Ogihara and Sharvit (2012)), how can ⌜*then*⌝ be compatible only with the deleted past? We know that ⌜*then*⌝ is compatibe with the deleted past, since ⌜*then*⌝ in English and Modern Greek is not only compatible with simple past-under-past sentences (which could in principle have a *de re* LF), but also with the Abusch-style examples with genuine deleted tense:

(48) A week ago, John said that in ten days he would say to his girlfriend that they **were** meeting **then** for the last time.

[7] Such constructions were first formulated in Abusch (1988) for English after Kamp and Rohrer's (1984) discussions on French.

(49) Prin mia evdhomadha, o Jorghos ipe oti se dheka meres tha eleghe stin
Before one week the Jorghos say.PAST that in ten days will say.IMPFV.PAST to-the
kopela tu oti **sinadjiondusan tote** ja teleftea fora.
girlfriend of-his that **meet.IMPFV.PAST then** for last time
'A week ago, Jorghos said that in ten days he would say to his girlfriend that they were meeting then for the last time' Modern Greek

Given that ⌜*then*⌝ is felicitous in the examples above, it is indeed compatible with deleted tense. But if this is the case, then the shifted present cannot be equivalent to deleted tense, or else there would be no ⌜*then*⌝-present puzzle. Thus, Vostrikova's (2019) analysis that ⌜*then*⌝ forces a *de re* reading of the embedded past will not work for English and Modern Greek.[8] Interestingly, in Russian too, *togda* 'then' does not always force a *de re* reading either, at least for some speakers who accept (50). Tsilia (2021) reports that two out of three consulted speakers accept the Abusch example in Russian with ⌜*then*⌝:

(50) %Nyedyelyoo nazad, Ivan skazal, čto čyeryez 10 dnyey on skazhyet svoyey dyevooshkye, čto
Week back, Ivan say.PAST that across 10 days he say.FUT his girlfriend that
oni **togda** vstryetilis' v poslyedniy raz.
they then meet.PAST.PFV in last time
'A week ago, Ivan said that in 10 days he would say to his girlfriend that they have then met for the last time' (Russian, Tsilia 2021: (119))

So, although the *de re* analysis could explain a subset of Russian speakers, it does not explain others and more importantly it does not explain the compatibility of ⌜*then*⌝ with a deleted past. Here, having expanded the ⌜*then*⌝-present puzzle under future, but also in languages with deleted tense, we show that unlike what has been assumed so far, tense deletion and tense shift should be treated differently (and LFs such as (42) should be set aside). In Section 6, we will show how to treat tense deletion and tense shift differently. We assume that tense shift is the result of perspective shift effected by **OP**$_\pi$. On the contrary, deleted tense has its perspective sensitivity precisely *deleted*, and is thus not affected by **OP**$_\pi$.

4 The compositional fragment

Having motivated the parametric account of tense shift, we now develop a compositional fragment of our system. As mentioned in the Introduction, our implementation draws heavily on theories of indexical shift,

[8] Vostrikova's theory also assumes *res*-movement (Heim, 1994) and a *de re together* constraint, according to which tenses and temporal adverbials have to be interpreted *de re* together. There is, however, evidence against this, since not all expressions within a single DP need to be interpreted *de re* together. The possibility of a 'mixed' reading has been reported in Romoli and Sudo (2009):

(1) *Context: Mary is watching television and sees Barack Obama, the actual president, and his sister besides him. Also, she doesn't know who he is and she thinks that the woman besides him must be is his wife.*
Mary thinks **the wife of the president** is nice. (Romoli and Sudo 2009: (20))

Here, *president* is read *de re*, while *wife* is read *de dicto*. But more importantly, Vostrikova's theory will not work for our data, since it cannot distinguish between shifted present and deleted past, which is the main goal of this paper.

especially Anand and Nevins (2004) and Anand (2006). The analysis of tense shift mirrors that of indexical shift (see also Schlenker (2003); Barry (2012) a.o. for similar analyses of tense). Both result from the shifting of the corresponding parameters, which in turn captures the **shift together** constraint observed in each domain. We briefly introduce the indexical shift formalism in Anand and Nevins, and then extend it to our theory of tense. Key differences between the two systems will be identified along the way.

4.1 Preliminaries: A theory of indexical shift

Indexical expressions such as *I, you, here, now, today, yesterday* can be shifted in environments like speech/attitude reports. This phenomenon is called indexical shift. Below is an example from Nez Perce:[9]

(51) Mipx Beth hi-neek--e [$\boxed{\text{pro}}$ kuu-se-∅]?
where.to Beth.NOM 3SUBJ-think-P-REM.PAST [1SG go-IMPERF-PRES]
✓Where did Beth$_i$ think I was going?
✓Where did Beth$_i$ think she$_i$ was going? (Nez Perce, Deal 2020: (5))

(51) is ambiguous between two possible interpretations of the embedded first-person pronoun. Under the non-shifted reading, the first-person pronoun draws its reference from the utterance context and refers to the speaker. Under the shifted reading, it draws its reference from a shifted context that represents the reported attitude, and refers to the author of that context, i.e. the attitude holder, Beth.[10]

Indexical shift exhibits the so-called **SHIFT TOGETHER** constraint. This constraint dictates that all shiftable indexicals (of the same type) within a minimal clausal domain must draw their references from the same context (see e.g. Anand 2006: (100)). This is illustrated by the Nez Perce example in (52), where the embedded clause contains two indexical pronouns:

(52) Ne-'níc-em pee-∅-n-e 'in-haama-na, [$\boxed{\text{'im-im}}$ ciq'aamqal
1SG-older.sister-ERG 3/3-say-P-REM.PAST 1SG-husband-ACC [2SG-GEN dog(ERG)
hi-twehkey'k-∅-e $\boxed{\text{'iin-e}}$].
3SUBJ-chase-P-REM.PAST 1SG-ACC]

a. ✓My sister$_s$ told my husband$_h$ that his$_h$ dog chased her$_s$. (both shifted)
b. ?My sister$_s$ told my husband$_h$ that your dog chased me. (both non-shifted)
c. ✗My sister$_s$ told my husband$_h$ that your dog chased her$_s$. (only the second shifted)
d. ✗My sister$_s$ told my husband$_h$ that his$_h$ dog chased me. (only the first shifted)

(Nez Perce, Deal 2020: (30))

[9]Languages that have been reported to allow indexical shift include e.g. Amharic (Schlenker, 1999), Zazaki (Anand and Nevins, 2004; Anand, 2006), Japanese (Sudo, 2012), Uyghur (Shklovsky and Sudo, 2014), Korean (Park, 2016), Nez Perce (Deal, 2020) etc.

[10]Note that the example contains overt *wh*-extraction from the embedded clause. This is to show that (51) involves genuine indexical shift within indirect discourse, since *wh*-extraction is impossible from a syntactically opaque direct quotation:

(1) *Who did Mary say, "I handed the bag to __"?
cf. Who did Mary say she handed the bag to __? (Deal 2020: ex. 11)

As indicated, only two out of the four possible readings are attested. The default interpretation (52a) is one where both indexical pronouns retrieve their references from the *reported* speech context, where the first-person pronoun *'iin-e* is interpreted as 'my sister' (the reported speaker) and the second-person pronoun *'im-im* as 'my husband' (the reported addressee). The other possible reading, albeit being less preferable, is one where neither indexical shifts as in (52b), i.e. the first-person pronoun refers to the speaker 'me' and the second-person pronoun to the addressee 'you'. Crucially, it is impossible for only one of the two embedded indexicals to shift, be it the first-person pronoun as in (52c) or the second-person pronoun as in (52d).

Anand and Nevins's analysis, which is a faithful incarnation of Kaplan's (1979) two-dimensional semantics, characterized the interpretation function as relative to a context parameter c and an evaluation index i, along with the function g for variable assignment:

(53) $[\![\cdot]\!]^{c,i,g}$

The context parameter c and the evaluation index i are treated as structurally 'homologous'. Here we will take both to be quadruples consisting of a speaker/author, an addressee, a time and a world:

(54) a. $c = \langle c_s, c_a, c_t, c_w \rangle$
b. $i = \langle i_s, i_a, i_t, i_w \rangle$

Indexical expressions draw their references directly from the relevant components of c, for instance:

(55) a. $[\![\text{I}]\!]^{c,i,g} = c_s$
b. $[\![\text{now}]\!]^{c,i,g} = c_t$

Speech/attitude predicates, whose complements are the usual habitat of shifted indexicals and shifted tense, are modeled as quantifiers over the indices used to evaluate the embedded clause. Take *say* as an example:

(56) $[\![\text{say}\ \varphi]\!]^{c,i,g} = \lambda x.\, \forall i' \in \textsc{say}(x)(i).\, [\![\varphi]\!]^{c,i',g}$
where $i' \in \textsc{say}(x)(i)$ just in case i' represents a speech context compatible with what x says in i.

The final ingredient of the analysis is the **context-shifting operator** $\textbf{OP}_c$ that overwrites the context parameter with the evaluation index, a Kaplanian 'monster':

(57) $[\![\textbf{OP}_c\ \varphi]\!]^{c,i,g} = [\![\varphi]\!]^{i,i,g}$

We illustrate this system with an example derivation of a shifted indexical, shown in (58). First the intensional predicate binds the index i' used to evaluate the reported speech (see (58i)). Second, $\textbf{OP}_c$ overwrites c with i' (see (58ii)), yielding the interpretation where the embedded indexical *me* refers to the speaker of the reported speech i'_s, i.e. Mark (see (58iii)). It is important to note that the index-binding predicate and the context-shifting operator go hand-in-hand in deriving indexical shift.

(58) $[\![\text{Mark says } \textbf{OP}_c \text{ Erica hates } \boxed{\text{me}}]\!]^{c,i,g} = 1$, iff

i. $\forall i' \in \textsc{say}(\textbf{m})(i_w).\ [\![\textbf{op}_c \text{ Erica hates } \boxed{\text{me}}]\!]^{c,i',g} = 1$, iff

ii. $\forall i' \in \textsc{say}(\textbf{m})(i_w).\ [\![\text{Erica is hates } \boxed{\text{me}}]\!]^{i',i',g} = 1$

iii. $\forall i' \in \textsc{say}(\textbf{m})(i_w).\ \textsc{hate}(\boxed{i'_s})(\textbf{e})(i'_w)$

The operator-based analysis accounts for the **Shift Together** constraint by assuming that $\textbf{op}_c$ can only be applied to propositional nodes. This is illustrated in (59). Either $\textbf{op}_c$ applies to the whole (embedded) clause as in (59b) and both indexicals draw their references from a shifted parameter, or $\textbf{op}_c$ does *not* apply and both indexicals are interpreted relative to the context of the utterance. No intermediate options are available.

(59) a. $[\![\text{your dog is chasing me}]\!]^{c,i,g} = \textsc{chase}(\textsc{dog-of}(\boxed{c_a}))(\boxed{c_s})(i_w)$

b. $[\![\textbf{op}_c \text{ your dog is chasing me}]\!]^{c,i,g} = [\![\text{your dog is chasing me}]\!]^{i,i,g}$
$= \textsc{chase}(\textsc{dog-of}(\boxed{i_a}))(\boxed{i_s})(i_w)$

Comparing (59) with the discussion in Section 2, it is clear that the perspective shifting operator behaves very similarly. To account for the ⌜*then*⌝-present puzzle, we need an analysis similar to the one provided for indexical shift, since the present and ⌜*then*⌝ should be anchored to the same perspective. Thus, our analysis will follow in spirit other shifted indexicality accounts of tense, such as Schlenker (2003) and Barry (2012). We will also treat the present as a shiftable indexical, and we will argue that, just like indexicals within a minimal domain are evaluated with respect to the same context, the present and ⌜*then*⌝ too are evaluated with respect to the same anchor. This is what is at the root of the ⌜*then*⌝-present puzzle. We will add the perspective parameter π and the corresponding shifting operator $\textbf{op}_\pi$ to our system in Sections 4.3 & 5. Before that, we enrich our compositional fragment to handle tense shift.

4.2 Preliminary: Compositions

Firstly, we would like to make a modification on the interpretation function $[\![\cdot]\!]$ for reasons independent from our empirical concerns. Instead of taking both the context c and index i as parameters, we will 'unload' the index i into the object language, as shown in (60). The place holder '…' in the array of parameters is reserved for the temporal perspective.

(60) $[\![\cdot]\!]^{c,\ldots,g} = \ldots\lambda i.\ldots$

This is motivated by the well-known problems of treating the index as a parameter (which Anand (2006): p. 108 already noted). Systems that treat the index as a parameter lack the necessary expressive power to capture a substantial fragment of natural language, unlike systems where evaluation indices are treated as variables (see e.g. Cresswell 1990). Allowing indices in the object language is also in better conceptual alignment with the pronominal treatment of tenses (Partee, 1973), which we adopt.

The price to pay for this modification is a slightly more complex compositional system that involves functional types corresponding to *intensional* properties and propositions. The system we adopt is essentially a variant of Gallin's (1975) Ty2, supplemented with a few idiosyncratic composition rules (adapted from Heim and Kratzer 1998) for simpler exposition.

The set of types τ is defined in (61). In addition to individuals $\mathtt{e}$ and truth values $\mathtt{t}$, the basic types also include worlds $\mathtt{s}$ and times $\mathtt{i}$. Derivative types include types of functions ($\tau{\rightarrow}\tau$) and products ($\tau{\times}\tau$).

(61) $\tau ::= \mathtt{e} \mid \mathtt{t} \mid \mathtt{s} \mid \mathtt{i} \mid \tau{\rightarrow}\tau \mid \tau{\times}\tau$

The product type $\tau{\times}\tau$ corresponds to ordered pairs of objects; and since either object may itself be a product type, it can be used to construct the types of tuples of any denumerable arity. In this paper, we will only use the product type for the type of the index i; since the index is now in the object language it needs a type. Like before, the index is a quadruple of speaker/author ($\mathtt{e}$), addressee ($\mathtt{e}$), time ($\mathtt{t}$) and world ($\mathtt{s}$). We will write $\mathtt{k}$ in short for the corresponding composite type:

(62) $\mathtt{k} ::= \mathtt{e}{\times}\mathtt{e}{\times}\mathtt{i}{\times}\mathtt{s}$

So κ is the type of the index i and of nothing else in this paper.

As usual, each basic type corresponds to a domain of objects in the model—$\mathcal{D}_{\mathtt{e}}$ is the domain of individuals, $\mathcal{D}_{\mathtt{t}} = \{0, 1\}$, $\mathcal{D}_{\mathtt{s}}$ is the set of worlds, and $\mathcal{D}_{\mathtt{i}}$ the domain of times, which we take to correspond to the set of closed time *intervals*. The domains corresponding to functional types and product types are derived as functional mappings of the Cartesian products of the more basic objects.

To illustrate this system, we have collected a list of relevant lexical entries in Table 3. Names and indexicals are of type $\mathtt{e}$, tenses and ⌜*then*⌝ of type $\mathtt{i}$. Common nouns have the type of intensional properties $\mathtt{k}{\rightarrow}\mathtt{e}{\rightarrow}\mathtt{t}$ (see for example *person*). This type essentially means "give me an index, then give me an individual, and I will give you a truth value" (compare with $\langle\kappa, \langle e, t\rangle\rangle$ in more standard notation). As for infinitives, they are intensional predicates requiring an additional time argument. Speech/attitude verbs like *say* have the type $\mathtt{k}{\rightarrow}\mathtt{i}{\rightarrow}(\mathtt{k}{\rightarrow}\mathtt{t}){\rightarrow}\mathtt{e}{\rightarrow}\mathtt{t}$, where $\mathtt{k}{\rightarrow}\mathtt{t}$ is the type of the proposition p (compare with $\langle\kappa, \langle i, \langle\langle\kappa, t\rangle, \langle e, t\rangle\rangle\rangle\rangle$). They essentially require an index ($\mathtt{k}$), a time ($\mathtt{i}$), a proposition ($\mathtt{k}{\rightarrow}\mathtt{t}$), an attitude holder ($\mathtt{e}$), and then they give a truth value ($\mathtt{t}$). Finally, quantifiers receive their usual extensional type $(\mathtt{e}{\rightarrow}\mathtt{t}){\rightarrow}(\mathtt{e}{\rightarrow}\mathtt{t}){\rightarrow}\mathtt{t}$ (compare with $\langle\langle e, t\rangle, \langle\langle e, t\rangle, t\rangle\rangle$). The semantics of tenses will be revised in Section 4.3.

lexical items	**types**	$[\![\cdot]\!]^{c,\dots,g}$
I/me	$\mathtt{e}$	c_s
Nate	$\mathtt{e}$	$\textbf{N}$
Erica	$\mathtt{e}$	$\textbf{E}$
person	$\mathtt{k}{\to}\mathtt{e}{\to}\mathtt{t}$	$\lambda i\,.\lambda x.\,\textbf{PERSON}(i)(x)$
be angry	$\mathtt{k}{\to}\mathtt{i}{\to}\mathtt{e}{\to}\mathtt{t}$	$\lambda i\,.\lambda t\,.\lambda x.\,\textbf{ANGRY}(i)(t)(x)$
hate	$\mathtt{k}{\to}\mathtt{i}{\to}\mathtt{e}{\to}\mathtt{e}{\to}\mathtt{t}$	$\lambda i\,.\lambda t\,.\lambda x\,.\lambda y.\,\textbf{HATE}(i)(t)(x)(y)$
say	$\mathtt{k}{\to}\mathtt{i}{\to}(\mathtt{k}{\to}\mathtt{t}){\to}\mathtt{e}{\to}\mathtt{t}$	$\lambda i\,.\lambda t\,.\lambda p\,.\lambda x.\,\forall i' \in \textbf{SAY}(i)(t)(x).\,p(i')$ (to be revised)
some	$(\mathtt{e}{\to}\mathtt{t}){\to}(\mathtt{e}{\to}\mathtt{t}){\to}\mathtt{t}$	$\lambda P\,.\lambda Q.\,\exists x.\,P(x)\,\wedge\,Q(x)$
every	$(\mathtt{e}{\to}\mathtt{t}){\to}(\mathtt{e}{\to}\mathtt{t}){\to}\mathtt{t}$	$\lambda P\,.\lambda Q.\,\forall x.\,P(x)\,\to\,Q(x)$
$\textbf{PRES}_n$	$\mathtt{i}$	$g(n)$ (to be revised)
$\textbf{PAST}_n$	$\mathtt{i}$	$g(n)$ (to be revised)

Table 3: list of lexical entries

As for composition rules, we have basic functional application (δ and σ are meta-variables for types):

(63) Functional Application (FA)

If α is a binary branching node with two children $\{\beta, \gamma\}$, where β is of type $\sigma \to \delta$ (i.e. $\beta : \sigma \to \delta$) and $\gamma : \sigma$, then $\alpha : \delta$ and $[\![\alpha]\!]^{c,\dots,g} = [\![\beta]\!]^{c,\dots,g}([\![\gamma]\!]^{c,\dots,g})$

We also make use of *intensional* functional application (IFA). There are three possible ways of applying IFA: (i) applying an intensional function to a non-intensional argument as in (64.1), (ii) applying a non-intensional function to an intensional argument as in (64.2), and (iii) applying an intensional function to an intensional argument as in (64.3). What makes a function or an argument intentional is dependency on the index (of type $\mathtt{k}$). All of them are simple extensions to FA, essentially percolating the index dependency from the child node(s) to the parent node. Notice that the parent node α is always of the type $\mathtt{k}\!\to \delta$.

(64) Intensional Functional Application (IFA)

If α is a binary branching node with two children $\{\beta, \gamma\}$:

1. if $\beta : \mathtt{k} \to \sigma \to \delta$ and $\gamma : \sigma$, then $\alpha : \mathtt{k} \to \delta$ and $[\![\alpha]\!]^{c,\dots,g} = \lambda i_{\mathtt{k}}.\,[\![\beta]\!]^{c,\dots,g}(i)([\![\gamma]\!]^{c,\dots,g})$
2. if $\beta : \sigma \to \delta$ and $\gamma : \mathtt{k} \to \sigma$, then $\alpha : \mathtt{k} \to \delta$ and $[\![\alpha]\!]^{c,\dots,g} = \lambda i_{\mathtt{k}}.\,[\![\beta]\!]^{c,\dots,g}([\![\gamma]\!]^{c,\dots,g}(i))$
3. if $\beta : \mathtt{k} \to \sigma \to \delta$ and $\gamma : \mathtt{k} \to \sigma$, then $\alpha : \mathtt{k} \to \delta$ and $[\![\alpha]\!]^{c,\dots,g} = \lambda i_{\mathtt{k}}.\,[\![\beta]\!]^{c,\dots,g}(i)([\![\gamma]\!]^{c,\dots,g}(i))$

We also use the following versions of **Predicate Modification** and **Predicate Abstraction** rules to deal with adjunction and quantification. As usual, Predicate Modification takes two intensional properties over objects of a type σ (e.g. individuals ($\sigma = \mathtt{e}$) or times ($\sigma = \mathtt{i}$)), and returns an intensional property of the same type conjoining them.[11] Predicate Abstraction takes an intensional proposition and a lambda binder (with a numbered index n), and returns a property that abstracts over a type σ variable.

[11] Recall that σ is just a meta-variable for types.

(65) Predicate Modification

If α is a binary branching node with two children $\{\beta, \gamma\}$, where $\beta : \texttt{k}{\rightarrow}\sigma{\rightarrow}\texttt{t}$ and $\gamma : \texttt{k}{\rightarrow}\sigma{\rightarrow}\texttt{t}$, then $\alpha : \texttt{k}{\rightarrow}\sigma{\rightarrow}\texttt{t}$ and $[\![\alpha]\!]^{c,\ldots,g} = \lambda i_{\texttt{k}}\, \lambda x_\sigma.\, [\![\beta]\!]^{c,\ldots,g}(i)(x) \wedge [\![\gamma]\!]^{c,\ldots,g}(i)(x)$.

(66) Predicate Abstraction

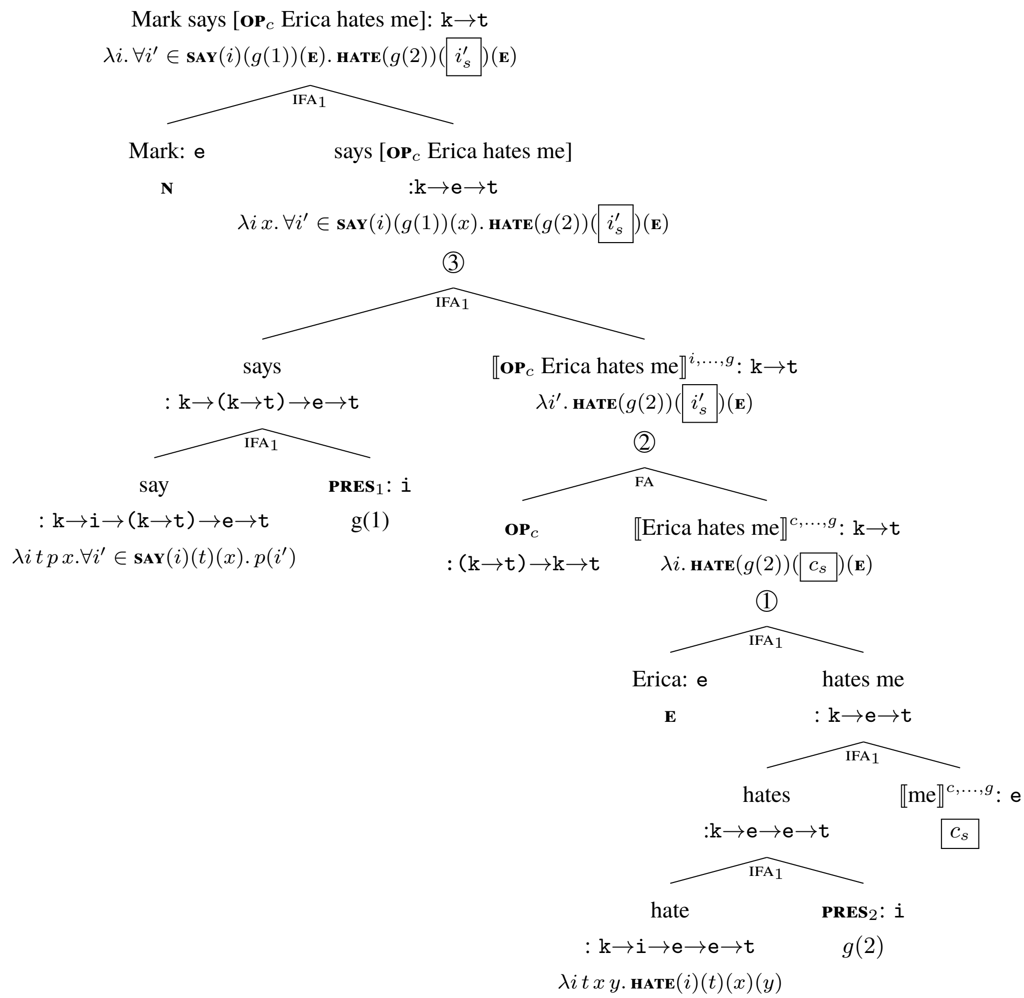
If α is a binary branching node with two children $\{\lambda_n, \beta\}$, where $\beta : \texttt{k}{\rightarrow}\texttt{t}$, then $\alpha : \texttt{k}{\rightarrow}\sigma{\rightarrow}\texttt{t}$ and $[\![\alpha]\!]^{c,\ldots,g} = \lambda i_k\, \lambda x_\sigma.\, [\![\beta]\!]^{c,\ldots,g[n\mapsto x]}$

We are now ready to illustrate how this compositional fragment works by deriving a sentence with indexical shift as we did for (58) before. The context-shifting operator $\textbf{OP}_c$ is defined as follows:

(67) $[\![\textbf{OP}_c\, \varphi]\!]^{c,\ldots,g} = \lambda i_{\texttt{k}}.\, [\![\varphi]\!]^{i,\ldots,g}(i)$

This is essentially the same as (57), where $\textbf{OP}_c$ substitutes the context parameter with the index parameter. Once again, $\textbf{OP}_c$ is assumed to only apply to propositional nodes (of type $\texttt{k}{\rightarrow}\texttt{t}$). The derivation of (58) proceeds as in (68). The composition rule implemented at each step is specified under each branching node.

(68) $[\![$Mark says $\textbf{OP}_c$ Erica hates $\boxed{\text{me}}\,]\!]^{c,\ldots,g}$



Note the key steps: ① is the result of computing the embedded proposition of type $\texttt{k}\to\texttt{t}$, where the first-person indexical is shown to draw its reference from the (local) context; ② is the result of applying $\textsc{op}_c$ to ①, thus overwriting the context parameter with the abstracted index i'; in ③ the speech predicate *say* binds the index, where the indexical corresponds to the attitude holder, Mark, rather than the speaker. The final result suggests that the sentence is true in c if and only if what Mark says at the time $g(1)$ overlapping with the time of utterance c_t is that Erica hates Mark. These are indeed the desired truth conditions.

4.3 Adding perspectives

We are now ready to integrate the perspective parameter to our compositional fragment. Recall that our theory of tenses mirrors that of indexicals in that tenses are interpreted relative to a temporal **perspective** parameter π, and tense shift is the result of perspectival shift. We further justify the addition of the perspective parameter in Section 7. For now, we simply add it to our system. The addition of π (where we had dots '…' before) completes the array of parameters of the interpretation function (cf. (60)):

(69) $[\![\cdot]\!]^{c,\pi,g} = ...\lambda i. ...$

Unlike the context parameter c and evaluation index i, which are both modeled as quadruples of type `k`, the perspective π is taken to be a simple time parameter (of type `i`) valued as a time interval. This is because in this paper we only need the temporal component of the perspective. It might turn out that perspectival sensitivity is needed in other domains too, in which case the perspective parameter could also become a quadruple. For now, we posit the simplest perspectival parameter possible, namely a time interval of type `i`. Tenses and ⌜*then*⌝ are treated as temporal pronouns with perspectival *presuppositions*, as shown in (70)–(72). Recall that '$\circ$' is read as 'temporally overlaps with', and '$<$' as 'temporally precedes'.

(70) $[\![\textbf{PRES}_n]\!]^{c,\pi,g} = g(n)$ only if $\boxed{g(n) \circ \pi}$, otherwise undefined.

(71) $[\![\textbf{PAST}_n]\!]^{c,\pi,g} = g(n)$ only if $\boxed{g(n) < \pi}$, otherwise undefined.

(72) $[\![\text{then}_n]\!]^{c,\pi,g} = g(n)$ only if $\boxed{\neg(g(n) \circ \pi)}$, otherwise undefined.

Other than the addition of π, everything is quite standard. **PRES**$_n$ presupposes that its reference $g(n)$ overlaps with the temporal perspective π (i.e. their intersection is non-empty), **PAST**$_n$ presupposes that $g(n)$ *precedes* π (i.e. all the time points in the interval $g(n)$ precede all time points in π), and then$_n$ is defined only if $g(n)$ is disjoint from π.[12][13]

Nothing significant hinges on the specific theory of presuppositions one assumes, so we can assume that they percolate along the derivations in a familiar fashion. For illustration, we will use the speech predicate *say*, treated as a presupposition filter, following Heim (1992):

[12]For now we set aside the potential complication that the perspective-sensitivity of ⌜*then*⌝ may be better characterized as an *anti-presupposition* (Percus 2006). This is shown in the following example (p.c. Philippe Schlenker):

(i) When it's sunny but cold, I don't go out; when it's sunny and warm, *then* I go out. So now I'm going out.

Then seems to refer to a time that at least includes the present, so the disjoint inference is not as stable as a presupposition might be. However, it seems that *then* here is restricted to the clause-initial position as in (i).

(ii) When it's sunny and warm, I (??then) go out (??then).

This suggests that *then* here is possibly a conditional *then* rather than a genuine temporal adverbial.

[13]A reviewer notes that mistaken belief cases, where the attitude holder misplaces themselves in time are hard to deal with in our system, since an extensional, contextually salient time cannot be anchored to a time in the attitude holder's temporal center. For example:

(1) At five, John thought.PAST_1 it was.PAST_2 seven.

We want the *seven* in the worlds compatible with John's beliefs to refer to the same time as the *five* in the actual world, without saying that tenses aren't rigidly fixed pronouns and without attributing a contradictory belief to John. We propose that tenses *are* rigidly fixed pronouns and rigidly fixed, but *five* and *seven* aren't. The latter are time concepts, i.e., functions from worlds to times, which do not rigidly refer across worlds. So, for all worlds w' compatible with John's beliefs in the actual world w_0, $[\![$ seven $]\!]^{c,\pi,g}$ $(w') = g(2) = g(1) =$ $[\![$ five $]\!]^{c,\pi,g}(w_0)$. Thus, *five* for example does not have the same reference in the worlds w' compatible with John's beliefs and in the actual world, i.e., $[\![$ five $]\!]^{c,\pi,g}(w') \neq [\![$ five $]\!]^{c,\pi,g}(w_0)$.

(73) $[\![\text{say}]\!]^{c,\pi,g} = \lambda i_{\texttt{k}}\,.\lambda t\,.\lambda p_{\texttt{k}\to\texttt{t}}\,.\lambda x.\,\forall i' \in \textsc{say}(i)(t)(x).\,[\![\varphi]\!]^{c,\pi,g}$

only if $\forall i' \in \textsc{say}(t)(x)(i_w), [\![\varphi]\!]^{c,\pi,g}(i')$ is defined, otherwise undefined

With that, here is a revised list of lexical entries, upgrading Table 3 with the semantics of tenses above:

lexical items	**types**	$[\![\cdot]\!]^{c,\pi,g}$
I/me	e	c_s
Nate	e	**N**
Erica	e	**E**
person	k→e→t	$\lambda i\,.\lambda x.\,\textbf{PERSON}(i)(x)$
be angry	k→i→e→t	$\lambda i\,.\lambda t\,.\lambda x.\,\textbf{ANGRY}(i)(t)(x)$
hate	k→i→e→e→t	$\lambda i\,.\lambda t\,.\lambda x\,.\lambda y.\,\textbf{HATE}(i)(t)(x)(y)$
say	k→i→(k→t)→e→t	$\lambda i\,.\lambda t\,.\lambda p\,.\lambda x.\,\forall i' \in \textsc{say}(i)(t)(x).\,p(i')$ only if $\forall i' \in \textsc{say}(t)(x)(i_w), [\![\varphi]\!]^{c,\pi,g}(i')$ is defined
some	(e→t)→(e→t)→t	$\lambda P\,.\lambda Q.\,\exists x.\,P(x) \wedge Q(x)$
every	(e→t)→(e→t)→t	$\lambda P\,.\lambda Q.\,\forall x.\,P(x) \to Q(x)$
$\textbf{PRES}_n$	i	$g(n)$ only if $g(n) \circ \pi$
$\textbf{PAST}_n$	i	$g(n)$ only if $g(n) < \pi$
then$_n$	i	$g(n)$ only if $\neg(g(n) \circ \pi)$

Table 4: list of lexical entries

Just like the context parameter c, the perspective parameter π may be shifted by an operator $\textsc{op}_\pi$ that applies only to propositional nodes of type k→t. The operator overwrites the perspective parameter of its prejacent with the time component of the evaluation index i_t:

(74) $[\![\textsc{op}_\pi\ \varphi]\!]^{c,\pi,g} = \lambda i_{\texttt{k}}.\,[\![\varphi]\!]^{c,i_t,g}(i)$

Finally, we have te following truth convention:

(75) Truth Convention

A sentence S with the LF φ is true relative to the utterance context c and assignment function g iff $[\![\varphi]\!]^{c,c_t,g}(c) = 1$.

What this truth condition essentially says is that the default value for the index is the context c, and the default value for the perspective is the temporal component of the context c_t. This will become relevant in relative clauses, where there is no intensional operator to shift the context and the perspective.

5 Solution to the ⌜*then*⌝-present puzzle

We now have all the necessary pieces to solve the puzzle. As already mentioned, tense shift in our account is achieved through the $\textsc{op}_\pi$ operator, which shifts the temporal perspective π. As for the incompatibility of $\ulcorner$*then*$\urcorner$ with the present, it follows from the conflicting presuppositions between them. More specifically, the present presupposes overlap with the perspective while $\ulcorner$*then*$\urcorner$ presupposes disjointness from it. In this Section, we show how this exactly works compositionally, as well as how we derive the tense-shift typology.

5.1 Root clauses

Let us start with the simplest sentence we need to account for, namely a matrix present with $\ulcorner$*then*$\urcorner$, as in (76):

(76) *Erica is angry then.

All relevant lexical entries have been given in Table 4. But how does the temporal adjunct $\ulcorner$*then*$\urcorner$ compose with the rest of the sentence? We assume that there is a silent preposition *during* before $\ulcorner$*then*$\urcorner$. When combined with *then*$_1$ below, *during* gives an intensional temporal property corresponding to the set of time intervals contained within *then*$_1$'s reference.

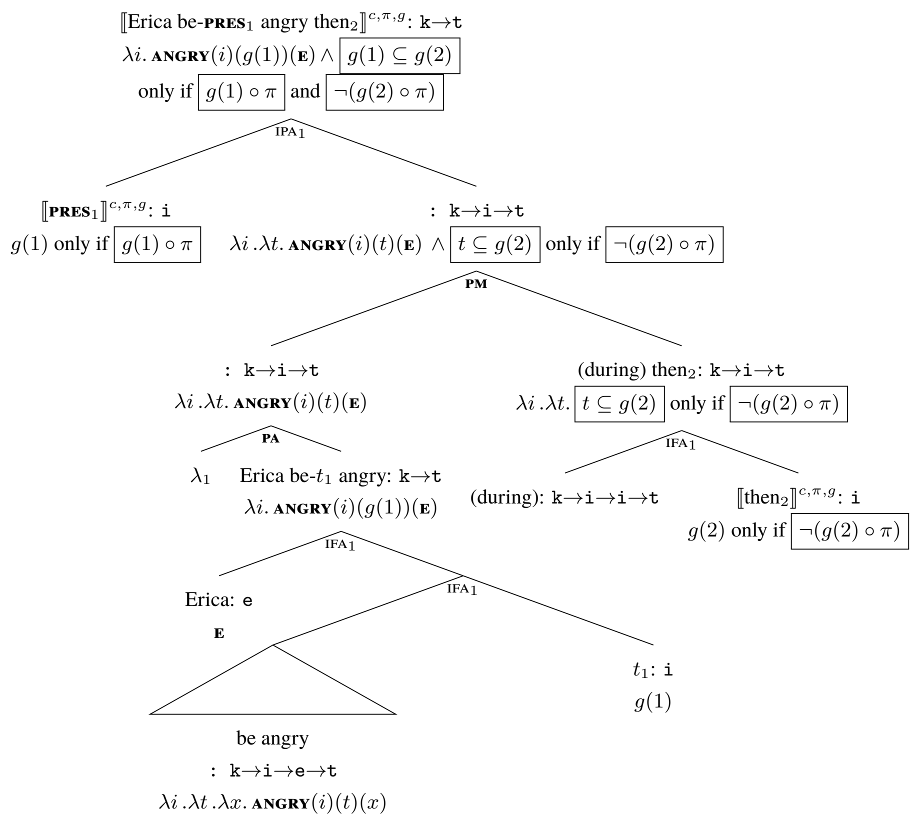
(77) a. $[\![\text{during}]\!]^{c,\pi,g} : \mathtt{k}\rightarrow\mathtt{i}\rightarrow\mathtt{i}\rightarrow\mathtt{t}$
$:= \lambda i_{\mathtt{k}} .\lambda t_{\mathtt{i}} .\lambda t'_{\mathtt{i}}. t' \subseteq t$

b. $[\![\text{then}_1]\!]^{c,\pi,g} = g(1)$ only if $\neg(g(1) \circ \pi)$

c. $[\![\text{during then}_1]\!]^{c,\pi,g} = \lambda i .\lambda t. t \subseteq g(1)$ only if $\neg(g(1) \circ \pi)$

The tense is quantifier-raised, so *during then* composes with the un-tensed VP via PREDICATE MODIFICATION (see (65)). Then, this new temporal property takes the present as an argument, as the LF in (78) shows. Remember that this is the predicted LF of an ungrammatical sentence.

(78) $[\![\textbf{PRES}_1\ \lambda_1\ [\text{Erica be-}t_1\text{ angry (during) then}_2]]\!]^{c,\pi,g}$



The reason the LF above is unattested is that the perspectival presuppositions of ⌜*then*⌝ and the present conflict with the assertion (as the boxes at the LF illustrate). We have (i) the perspectival presupposition of PRES$_1$, i.e. $g(1) \circ \pi$, (ii) the perspectival presupposition of then$_2$, i.e. $\neg(g(2) \circ \pi)$, and (iii) the temporal restriction $g(1) \subseteq g(2)$. These are conflicting conditions: if (i) and (ii) are satisfied, the assertion cannot hold. This is because the part of $g(1)$ that overlaps with π cannot be contained within $g(2)$, since the latter is disjoint from π. The ⌜*then*⌝-present incompatibility thus follows as a semantic anomaly—there is no context in which a sentence like (76) is true with its presuppositions satisfied.

5.2 Shifted present in past speech/attitude reports

How about the incompatibility of ⌜*then*⌝ with a shifted present? Recall that this was the case for Modern Greek, Modern Hebrew, Russian and Japanese, all languages with a shiftable present in speech/attitude reports. Using English pseudo-glossing, these constructions correspond to the following LF:

(79) *Nate said Erica is angry then.

⟦Nate say-PAST$_1$ [$\boxed{\textbf{OP}_\pi}$ Erica be-PRES$_2$ angry then$_3$]⟧$^{c,\pi,g}$

The derivation for (79) is in (80). The tense in the embedded clause shifts similar to how indexicals shift in (68). There are two crucial steps: at step ①, $\textsc{op}_\pi$ applies, overwriting the temporal perspective with an abstract temporal index i'_t. Since the operator shifts the perspective of the whole embedded clause, $\textsc{pres}_2$ and then$_3$ shift together; at step ②, the intensional predicate *say* binds the evaluation index i' of the embedded clause, associating the time component i'_t with the time of the reported speech. The embedded $\textsc{pres}_2$ ends up denoting a time overlapping with the time of the attitude $g(1)$, while *then*$_3$ a time disjoint from it. Like before, a conflict arises between the perspectival presuppositions of $\textsc{pres}_2$ and *then*$_3$ and the asserted temporal restriction $g(2) \subseteq g(3)$. This explains why ⌜*then*⌝ cannot be used to restrict the shifted present.

(80) $[\![$Nate say-$\textsc{past}_1$ [$\boxed{\textsc{op}_\pi}$ Erica be-$\textsc{pres}_2$ angry then$_3$]$]\!]^{c,\pi,g}$

$[\![$Nate say-$\textsc{past}_1$ [$\textsc{op}_\pi$ Erica be-$\textsc{pres}_2$ angry then$_3$]$]\!]^{c,\pi,g}$: `k→t`
$\lambda i.\, \forall i' \in \textsc{say}(i)(g(1))(\textsc{n}).\, \textsc{angry}(i)(g(2))(\textsc{e}) \wedge \boxed{g(2) \subseteq g(3)}$
only if for all $i' \in \textsc{say}(i)(g(1))(\textsc{n})$, $\boxed{g(2) \circ i'_t (= g(1))}$ and $\boxed{\neg(g(3) \circ i'_t (= g(1)))}$

IFA$_1$

Nate: `e`
$\textsc{n}$

said [$\textsc{op}_\pi$ Erica be-$\textsc{pres}_2$ angry then$_3$] : `k→e→t`
$\lambda i\,.\lambda x.\, \forall i' \in \textsc{say}(i)(g(1))(x).\, \textsc{angry}(i)(g(2))(\textsc{e}) \wedge \boxed{g(2) \subseteq g(3)}$
only if for all $i' \in \textsc{say}(i)(g(1))(x)$, $\boxed{g(2) \circ i'_t}$ and $\boxed{\neg(g(3) \circ i'_t)}$
②

IFA$_1$

said$_1$: `k→(k→t)→e→t`

IFA$_1$

say
: `k→i→(k→t)→e→t`
$\lambda i\,.\lambda t\,.\lambda p\,.\lambda x.\forall i' \in \textsc{say}(i)(t)(x).\, p(i')$

$[\![\textsc{past}_1]\!]^{c,\pi,g}$: `i`
$g(1)$ only if $g(1) < \pi$

$[\![\textsc{op}_\pi$ Erica be-$\textsc{pres}_2$ angry then$_3]\!]^{c,\pi,g}$: `k→t`
$\lambda i'.\, \textsc{angry}(i')(g(2))(\textsc{e}) \wedge \boxed{g(2) \subseteq g(3)}$
only if $\boxed{g(2) \circ i'_t}$ and $\boxed{\neg(g(3) \circ i'_t)}$
①

FA

$\textsc{op}_\pi$
: `(k→t)→k→t`

$[\![$Erica be-$\textsc{pres}_2$ angry then$_3]\!]^{c,\pi,g}$: `k→t`
$\lambda i.\, \textsc{angry}(i)(g(2))(\textsc{e}) \wedge \boxed{g(2) \subseteq g(3)}$
only if $\boxed{g(2) \circ \pi}$ and $\boxed{\neg(g(3) \circ \pi)}$ (= (78))

Note again the similarity with indexical shift. Steps ① and ② go hand in hand, since an intensional operator, such as an attitude verb, is needed to bind the index i' that $\textsc{op}_\pi$ introduced. Otherwise, without *say* the dependency on i' would percolate all the way to the top, since nothing would bind i'. This is also where the TRUTH CONVENTION in (75) becomes relevant. Had there been no intensional operator like *say* to bind the index, the truth convention would be used instead. The latter states that the default value for i' is the utterance context; thus, without *say* both the embedded present and *then* would be anchored to the time of utterance. This is why, in languages such as Modern Greek, Modern Hebrew and Russian, tense shift is unavailable in relative clauses. Because they are extensional environments, and there is no intensional operator to bind the index.[14] There is only one exception to this, as we saw in Section 2, namely Japanese. In Japanese, surprisingly, the present can also shift in relative clauses, despite the absence of an intensional operator. This is a puzzle that is entirely independent from ours, and thus we will not attempt to solve it. However, there is a way of technically handling these cases in our framework, namely by assuming that Japanese tenses can bind indices, like intensional verbs. We derive the relevant examples under this assumption in Appendix A.

So far, we have shown how ⌜*then*⌝ and the present have conflicting presuppositions, whenever the present shifts. Another environment where the present shifts is under a future modal. We explore this next.

5.3 Shifted present under future

As we saw in Section 2, all languages surveyed in this paper allow for the present to shift under future, both in attitude reports as well as in relative clauses. The reason the present can shift under future both in intensional and extensional environments is the semantics of the future. It is the intensional future operator (*will* in English) that binds the evaluation index, thus shifting the present. In fact, the future operator can bind all components of the evaluation index, not just the temporal component.

We decompose ⌜*will*⌝ (quotes indicate it is a placeholder for all future operators) into an index-binding operator WOLL and a present tense, as shown in (81). WOLL is a modal, because it has an 'accessibility' function ACC, which takes an index i and a time argument t and quantifies over all accessible indices i'. The future reading is the result of $i'_t > t$.[15] In addition to the index and the time, WOLL also takes a temporal property p as an argument. The latter is of type $\mathtt{k}{\to}\mathtt{i}{\to}\mathtt{t}$, taking an index and a time, and returning a truth value. So, p takes the index i' and its temporal component i'_t as arguments, yielding a truth value.

(81) $[\![will_n]\!]^{c,\pi,g} = [\![\textsc{woll-pres}_n]\!]^{c,\pi,g} : \mathtt{k}{\to}(\mathtt{k}{\to}\mathtt{i}{\to}\mathtt{t}){\to}\mathtt{t}$, where

$$[\![\textsc{woll}]\!]^{c,\pi,g} : \mathtt{k}{\to}\mathtt{i}{\to}(\mathtt{k}{\to}\mathtt{i}{\to}\mathtt{t}){\to}\mathtt{t}$$
$$:= \lambda i\,.\lambda t\,.\lambda p_{\mathtt{k}\to\mathtt{i}\to\mathtt{t}}.\,\forall i' \in \textsc{acc}(i)(t).\, i'_t > t \wedge [\![p]\!]^{c,\pi,g}(i')(i'_t)$$
only if $[\![p]\!]^{c,\pi,g}(i')(i'_t)$ is defined for all $i' \in \textsc{acc}(i)(t)$, otherwise undefined

To illustrate how this works, we derive a speech report under future in (82) as well as a relative clause under future in (83). In both examples, $\textsc{op}_\pi$ applies to the embedded proposition, shifting the perspective to the time component of an abstract index. In (82), the index is bound by the the intensional predicate *say*, whose time argument is supplied by *will*. Therefore, the perspective of the shifted present and ⌜*then*⌝ are indirectly bound by *will*. In (83), on the other hand, the main verb *meet* is extensional and thus cannot bind the index; the shifted perspective is thus directly bound by *will*. In both cases, ⌜*then*⌝ and the present are anchored to i'_t, introduced by WOLL. Like before, we have a conflict between their presuppositions and the assertion, resulting to semantic anomaly.

[14] This does not hold in relative clauses under the future, since the future itself is modal, serving as the necessary intensional operator.

[15] Whether the future reading is part of WOLL or a prospective aspect it combines with is beyond the scope of this paper. We also do not delve deeper into the modal force of WOLL. These are interesting debates about the semantics of the future, which nonetheless fall outside the scope of the paper.

(82) *Nate will$_0$ say Erica is$_2$ angry then$_3$

$[\![$ **woll-pres**$_0$ λ_1 [Nate say-t_1 [**op**$_\pi$ Erica is-**pres**$_2$ angry then$_3$]]$]\!]^{c,\pi,g}$

$$\lambda i.\, \forall i' \in \textsc{acc}(i)(g(0)).\, i'_t > g(0) \;\wedge\; \forall i'' \in \textsc{say}(i')(i'_t)(\textsc{n}).\, \textsc{angry}(i)(g(2))(\textsc{e}) \wedge \boxed{g(2) \subseteq g(3)}$$
only if $g(0) \circ \pi$ and for all $i' \in \textsc{acc}(i)(g(0))$, $\boxed{g(2) \circ i'_t}$ and $\boxed{\neg(g(3) \circ i'_t)}$

will$_0$: $\mathtt{k}\to(\mathtt{k}\to\mathtt{i}\to\mathtt{t})\to\mathtt{t}$: $\mathtt{k}\to\mathtt{i}\to\mathtt{t}$

IFA$_1$ PA

woll: $\mathtt{k}\to\mathtt{i}\to(\mathtt{k}\to\mathtt{i}\to\mathtt{t})\to\mathtt{t}$ **pres**$_0$: $\mathtt{i}$ λ_1

$[\![$Nate say-t_1 [**op**$_\pi$ Erica be-**pres**$_2$ angry then$_3$]$]\!]^{c,\pi,g}$: $\mathtt{k}\to\mathtt{t}$

$\lambda i.\, \forall i' \in \textsc{say}(i)(g(1))(\textsc{n}).\, \textsc{angry}(i)(g(2))(\textsc{e}) \wedge \boxed{g(2) \subseteq g(3)}$

only if for all $i' \in \textsc{say}(i)(g(1))(\textsc{n})$, $\boxed{g(2) \circ g(1)}$ and $\boxed{\neg(g(3) \circ g(1))}$

(83) *Nate will meet someone who is angry then$_5$

$[\![$ **woll-pres**$_0$ λ_1 [some (**person**) (λ_2 [**op**$_\pi$[x_2 be-**pres**$_3$ angry then$_5$]])]λ_6 [Nate meet-t_1 x_6]$]\!]^{c,\pi,g}$

$$\lambda i.\, \forall i' \in \textsc{acc}(i)(g(0)).\, i'_t > g(0) \;\wedge\; \exists x.\, \textsc{person}(i')(x) \;\wedge \textsc{angry}(i')(g(3))(x) \;\wedge\; \textsc{meet}(i')(i'_t)(x)(\textsc{n}) \;\wedge\; \boxed{g(3) \subseteq g(5)}$$
only if $g(0) \circ \pi$ and for all $i' \in \textsc{acc}(i)(g(0))$ $\boxed{g(3) \circ i'_t}$ and $\boxed{\neg(g(5) \circ i'_t)}$

IFA$_1$

will$_0$: $\mathtt{k}\to(\mathtt{k}\to\mathtt{i}\to\mathtt{t})\to\mathtt{t}$:$\mathtt{k}\to\mathtt{i}\to\mathtt{t}$

IFA$_1$ PA

woll: $\mathtt{k}\to\mathtt{i}\to(\mathtt{k}\to\mathtt{i}\to\mathtt{t})\to\mathtt{t}$ **pres**$_0$: $\mathtt{i}$ λ_1

$[\![$Nate meet-t_1 someone who **op**$_\pi$ is$_3$ angry then$_5$$]\!]^{c,\pi,g}$:$\mathtt{k}\to\mathtt{t}$

$\lambda i.\, \exists x.\, \textsc{person}(i)(x) \;\wedge \textsc{angry}(i)(g(3))(x) \wedge$

$\textsc{meet}(i)(g(1))(x)(\textsc{n}) \;\wedge\; \boxed{g(3) \subseteq g(5)}$

only if $\boxed{g(3) \circ i_t}$ and $\boxed{\neg(g(5) \circ i_t)}$

This explains the incompatibility of ⌜*then*⌝ with the present, assuming the latter is interpreted. But what if it is deleted instead? Indeed, the sequence of tense rule in (44), repeated below as (84), applies to any tense morpheme c-commanded by an agreeing tense morpheme. So, as a reviewer points out, if **will** is decomposed into **woll** and **pres**, the embedded present should have a deleted LF in SOT languages, and this deleted LF should give rise to a coherent reading. For example, (82) should have an alternative LF where the embedded present is deleted, thus not giving rise to the puzzle. In the next Section, we will see how deleted tenses escape the ⌜*then*⌝-present puzzle by not being sensitive to the perspective. There, we will discuss the possibility of the deleted present-under-future LF in (85). We will argue that this deleted LF is precisely why *then* is compatible with the present under future in English, an SOT language. We will also argue that this LF is unavailable in Modern Greek, since it has a *Prefer to Shift* (rather than delete) rule.

(84) Sequence-of-tense (SoT)
The tense feature of a tense c-commanded by an agreeing tense may be deleted at the Logical Form.

(85) $[\![\textsc{woll-pres}_0\ \lambda_1$ [Nate say-t_1 [$\textsc{op}_\pi$ Erica be-$\cancel{\text{pres}}_2$ angry then$_3$]]$]\!]^{c,\pi,g}$

5.4 Interim summary

In this Section we showed what goes wrong compositionally when ⌜*then*⌝ and the present appear in the same clause. Irrespective of the language or the embedding environment where the puzzle occurs, the causes are the same. We saw that ⌜*then*⌝ cannot restrict the reference of the present when they share the same temporal perspective. This follows directly from our parametric approach, where the perspective is a parameter and thus has to be shared by ⌜*then*⌝ and the present when they appear in the same minimal clausal domain.

We implemented tense shift as the result of a propositional operator $\textsc{op}_\pi$ shifting the perspective. What $\textsc{op}_\pi$ does is overwrite the perspective parameter introducing an abstract evaluation index. That index is then bound by a higher intensional operator (be it an attitude verb or a future or, for Japanese, any tense as shown in Appendix A). The bound index is thus distinct from the time of utterance. In our system, all tense shift occurs via this operator $\textsc{op}_\pi$.

We also saw that in some languages where the puzzle occurs, like Modern Greek, Modern Hebrew and Russian, only intensional operators such as a attitude verbs or a future operator bind indices. Thus, tense shift is restricted to these environments. As for Japanese, tense shift occurs in seemingly extensional environments, and we propose in Appendix A that Japanese tenses are themselves intensional and thus index-binders facilitating tense shift.[16]

6 Deleted tenses are not perspective-sensitive

A crucial point of this paper is the asymmetry between *shifted* present and *deleted* tense. Recall that in Modern Greek, a language with both a shifted present and a deleted past (Sharvit, 2018; Tsilia, 2021, 2022a,b), ⌜*then*⌝ is only compatible with the latter. This has theoretical importance, since the shifted present and the deleted past were up to now assumed to be semantically equivalent (Ogihara and Sharvit, 2012). The puzzle arising in Modern Greek only with the shifted present, but not with the deleted past, teaches us that they cannot be equivalent. In this Section we propose to model their difference in terms of sensitivity to the perspective parameter: the shifted present is sensitive to the perspective, but the deleted past is not.

[16] An alternative analysis would be to take the behavior of Japanese as a baseline, and assume all tenses (not just Japanese ones) are intensional. However, we would then over-generate readings for Modern Greek, Modern Hebrew, English and Russian. This is why we opted for this analysis which ties intensionality and tense shift.

The key diagnostic for deleted tense, (48), is repeated below as (86). Recall that this example shows that the most deeply embedded past has to be deleted in English, since it is not past relative to any other moment in the sentence. Notice, also, that it is compatible with *then*:

(86) A week ago, John said that in ten days he would say to his girlfriend that they **were** meeting **then** for the last time.

Our system provides a straightforward explanation for the contrast. Unlike the shifted present, we argue that deleted tenses are plain temporal pronouns without any perspectival presupposition:

(87) $[\![\cancel{\textbf{PRES}}_n]\!]^{c,\pi,g} = g(n)$

(88) $[\![\cancel{\textbf{PAST}}_n]\!]^{c,\pi,g} = g(n)$

Let us illustrate this with the following working example (which has a structure similar to (86)). We consider two possible LFs, with and without the perspective-shifting operator.

(89) John thought$_1$ a week ago that in ten days Mary would$_1$ say that she **missed**$_2$ Nate **then**$_3$.

a. John think-**PAST**$_1$ a week ago
[**WOLL**-~~**PAST**~~$_1$ λ_2 [Mary say-t_2 in ten days
[Mary miss-~~**PAST**~~$_2$ Nate then$_3$]]]

b. John think-**PAST**$_1$ a week ago
[**WOLL**-~~**PAST**~~$_1$ λ_2 [Mary say-t_2 in ten days
OP$_\pi$ [Mary miss-~~**PAST**~~$_2$ Nate then$_3$]]]

We focus on (89a) and (89b) for a minimal comparison. As usual, *would* is simply the past tense of *will*, and is thus decomposed into WOLL and a past tense **PAST**$_1$. In both LFs ~~**PAST**~~$_1$ as well as ~~**PAST**~~$_2$ are deleted.[17] The only 'real' past tense is the matrix past on the attitude verb.

The denotation of the most deeply embedded clause without **OP**$_\pi$ in (89a) is given in (90a) below. Crucially, as shown in the lexical entries above, deleted tense does not carry any presupposition—it is precisely *deleted*. Thus, only ⌜*then*⌝ carries its usual presupposition, namely that its reference is disjoint from the temporal perspective. If we take the LF with **OP**$_\pi$, as in (89b), with the denotation in (90b) below, ⌜*then*⌝ presupposes that its reference is disjoint from the index that **OP**$_\pi$ introduced (rather than the perspective as in (89a)). Importantly, **OP**$_\pi$ does not affect at all the interpretation of ~~**PAST**~~$_2$. As such, in both cases there is no conflict between deleted tense and ⌜*then*⌝.

(90) a. $[\![$Mary miss-~~**PAST**~~$_2$ Nate then$_3]\!]^{c,\pi,g}$
$= \lambda i.\, \textbf{MISS}(i)(g(2))(\textbf{M})(\textbf{N}) \wedge \boxed{g(2) \subseteq g(3)}$, only if $\boxed{\neg(g(3) \circ \pi)}$

[17] For the past tense on *would*, we could have assumed that is interpreted *de re*, relative to the time of the utterance, as opposed to deleted; however, this option is not available for the past tense on *miss*. As shown in Section 3, the most deeply embedded past *has to* be deleted. The one on *would* does not *have to* be deleted, but it *can*. Here, we go for the latter, although this choice will not matter.

b. $[\![\textsc{op}_\pi$ [Mary miss-~~PAST~~$_2$ Nate then$_3$]$]\!]^{c,\pi,g}$
$= \lambda i'.\, [\![$Mary miss-~~PAST~~$_2$ Nate then$_3]\!]^{c,i',g}$
$= \lambda i'.\, \textsc{miss}(i')(g(2))(\textsc{m})(\textsc{n}) \wedge \boxed{g(2) \subseteq g(3)}$ only if $\boxed{\neg(g(3) \circ i'_t)}$

Indeed, if we compare (90) to the semantic anomaly triggered by ⌜*then*⌝ and the present (as in (78), (82) and (83)), there is one less presupposition. More specifically, the presupposition of the shifted present is missing, and thus the semantic anomaly is prevented. This is the case both with the LF with and without $\textsc{op}_\pi$, since the only presupposition in both cases is the one carried by *then*$_3$, which does not contradict the asserted temporal restriction $g(2) \subseteq g(3)$. Thus, both LFs lead to semantically coherent meanings.

So, what is the difference between the two LFs, with and without $\textsc{op}_\pi$? The difference lies in whether the local perspective is shifted by $\textsc{op}_\pi$ or not. If it shifted to the time of Mary's speech, the presupposition of *then*$_3$ will dictate that *then* denotes a time disjoint from the time of Mary's speech. Thus, *then* cannot denote a time overlapping Mary's speech and we cannot get the simultaneous reading. Instead, the simultaneous reading in (86) is obtained through the LF in (90a), without the perspective shifting operator. In (90a), the presupposition of *then*$_3$ simply requires that it is disjoint from the time of utterance, which is satisfied. Thus, the reference of *then*$_3$ can contain the time of Mary's speech in three days from now, and we get the simultaneous reading.

As mentioned in the last Section, the deleted past is not the only possible deleted tense. The SOT rule (see (84) above) applies to any tense morpheme c-commanded by an agreeing tense morpheme, and as such it also applies to the present under **WILL** (decomposed into **WOLL** and **PRES**). As a reviewer points out, the possibility of a deleted present under future explains the compatibility of *then* with present under future in English in (31d) and (32d), repeated below as (91) and (92) respectively:

(91) In 2030, John will say that he is in jail ($^{\%}$**then**). (Tsilia 2021: (85))

(92) In 2030, John will be married to someone who is in jail (**then**).
(Tsilia 2021: (87), adapted from von Fintel and Heim (1997-2021): §7, (4))

The examples above escape the ⌜*then*⌝-present puzzle, since the embedded present is deleted.[18] The deleted present has the entry in (87), and the LFs of the English examples are below:

(93) $[\![$**WOLL-PRES**$_0$ λ_1 [John say-t_1 [$\textsc{op}_\pi$ he be-~~pres~~$_2$ in-jail then$_3$]]$]\!]^{c,\pi,g}$

(94) $[\![$**WOLL-PRES**$_0$ λ_1 [some (**PERSON**) (λ_2 [$\textsc{op}_\pi$[x_2 be-~~pres~~$_3$ in-jail then$_5$]])]λ_6 [John married-t_1 x_6]$]\!]^{c,\pi,g}$

[18]As the same anonymous reviewer points out, there is still a puzzle with present-under-*would* readings (Anand and Hacquard, 2008; Wurmbrand, 2014; Sharvit, 2014) being compatible with *then*, such as the following (tested with 3 speakers):

(1) Context: A week ago I saw John and he was very concerned about his relationship. He said he would meet his girlfriend in 10 days to break up. He specifically said: "In 10 days I'll tell her that this is the last time we're meeting".
A week ago, John said that in ten days he would say to his girlfriend that they are meeting then for the last time.

This is puzzling since *would* is composed of WOLL and PAST, and thus the most embedded present does not get deleted. We leave a more thorough investigation of such examples for future research, but we point out that the present here could actually be interpreted as a future. In other words, the speaker could be taking John to have said "I am breaking up with my girlfriend then/in 10 days".

But if English has these LFs which escape the ⌜*then*⌝-present puzzle under future, then Modern Greek, another SOT language, should also have a deleted present under ⌜*will*⌝. Yet, in subsection 5.3 above, we explained the incompatibility of ⌜*then*⌝ with the present under future in Modern Greek as an instance of the puzzle. Crucial to this argument was the idea that the present is interpreted, and is thus perspective-sensitive, conflicting with the perspectival presupposition of ⌜*then*⌝. If a deleted present is allowed in English, what blocks it in Modern Greek? In other words, why does Modern Greek pattern with Modern Hebrew and Russian rather than English when it comes to present-under-future?

We believe the answer lies in the fact that the Modern Greek present is shiftable across the board, like in Modern Hebrew and Russian, and unlike in English. More specifically, we propose that the Modern Greek present has to shift rather than get deleted under future. This is achieved through a *Prefer to Shift* rule, which prioritizes a shiftable over a deleted present, when the language has both.[19] Thus, the Modern Greek present ends up shifting rather than getting deleted under future, which explains its incompatibility with ⌜*then*⌝ (see 5.3). As for English, there are two possibilities. Since the English present cannot generally shift, English could exclusively have a deleted LF for present-under-future, which explains the felicity of *then* in the construction. Another possibility is that the English present can shift under future, but English lacks the *Prefer to Shift* rule. We leave a full exploration of this issue for future research.

Having shown that ⌜*then*⌝ is compatible with deleted tense, and having cashed that out in our system in terms of deleted tenses being perspective insensitive, we will now present evidence that the perspective parameter is not redundant, but must indeed be distinct from both the context and the index.

7 Perspective is not Context or Index

So far, we have accounted for the data discussed in Sections 2 and 3 by arguing that, contrary to what has been assumed up to now, deleted and shifted tense do not have the same interpretation. Instead, we proposed that although shifted present is sensitive to the perspective parameter π, deleted tenses are not. One might be wondering at this point: what exactly is this perspective parameter?

In this Section, we will provide a partial answer by pointing out what the perspective is *not*. In the interest of theoretical simplicity, one may be tempted to identify the perspective with (the temporal component of) other interpretation parameters, such as the context or the index. We argue that this cannot be the case, and thus the perspective has to be a *distinct* parameter, especially designated for the interpretation of tenses.

7.1 Perspective is not Context

Suppose, for the sake of contradiction, that π is in fact just the time component c_t of the context parameter. Then, tenses and ⌜*then*⌝ are essentially temporal indexicals like ⌜*now*⌝, ⌜*tomorrow*⌝ and ⌜*yesterday*⌝ whose

[19] Notice that *Prefer to Shift* does not rule out an indexical present, since it is not a *deleted* present. In a sense, *Prefer to Shift* prioritizes an LF where every tense is interpreted over one where there is a deleted tense. Compare with other similar rules, such as *Prefer De Se* (Schlenker, 1999; Ogihara and Sharvit, 2012; Tsilia, 2021, 2022a,b), which prioritizes *de se* over *de re* LFs.

interpretations are dependent on the utterance context, as illustrated below:

(95) a. $[\![\textbf{PRES}_n]\!]^{c,g} = g(n)$ only if $\boxed{g(n) \circ c_t}$, otherwise undefined.

b. $[\![\text{then}_n]\!]^{c,g} = g(n)$ only if $\boxed{\neg(g(n) \circ c_t)}$, otherwise undefined.

c. $[\![\text{now}]\!]^{c,g} = c_t$

In this case, assuming that tense shift is due to the shift of c_t makes the additional prediction that languages that allow tense shift would allow for concomitant indexical shift as well. This is not the case, see e.g. the example below from Modern Greek:

(96) To 1960, o Yanis iksere oti i Maria **ine** omorfi (#**tora**).
DET 1960, DET Yanis know.PAST that DET Maria **be.PRES** beautiful **now**
'In 1960, Yanis knew that Maria was (LIT.: is) beautiful.' Modern Greek (Tsilia 2021: (100))

Without the temporal indexical *tora* 'now', the embedded present in (96) can receive a shifted reading where it reports Yanis' past knowledge in 1960 that Maria was beautiful at that time. However, adding the temporal indexical *tora* 'now' forces a rather marginal reading that Yanis clairvoyantly predicted Maria's state of beauty at the time of utterance. In other words, the temporal indexical *tora* 'now' does not shift, has to refer to the time of utterance, and therefore blocks the shifted interpretation of the embedded present. The same holds for Modern Hebrew and Japanese, as shown in (97) and (98):

(97) Lifney alpayim šana, Yosef **xašav** še (#**axšav**) Miriam **ohevet** oto.
before two-thousand year Yosef **think.PAST** that **now** Miriam **love.PRES** him
'Two thousand years ago, Yosef thought that Miriam loved (LIT.: loves) him.' Modern Hebrew

(98) ninenn mae, Yusuke-wa Yukiko-ga (#**ima**) ninnshinn **shite-iru** to **shitteita**
2-years before Yusuke-TOP Yukiko-NOM **now** pregnant **be-PRES** with **know.PST**.
'2 years ago, Yusuke knew that Yukiko was (LIT.: is) pregnant.' Japanese

Relatedly, while indexical shift is generally recognized as a phenomenon restricted to speech/attitude reports (e.g. both Anand (2006) and Deal (2020) propose a shifting operator *selected* by certain speech/attitude predicates), tense shift, as we have seen, may occur in non-attitudinal environments as well, in languages like Japanese. Therefore even if the underlying mechanisms are similar, tense shift and indexical shift should not follow from the exact same operation. We take these to be strong evidence for distinguishing π and c_t.

That being said, we would like to point out that separating π and c_t does not mean that they are completely independent through the course of semantic evaluations, and their respective values may be correlated in a systematic way. We will return to this point shortly.

7.2 Perspective is not Index

Now suppose, for the sake of contradiction, that the perspective is just the temporal component of the evaluation index, i.e. $\pi = i_t$. This is roughly the position taken by many previous approaches (e.g. Ogihara 1996;

Abusch 1997; Kratzer 1998; Kusumoto 1999; Ogihara and Sharvit 2012 a.o.). In that case, the semantics of tenses and ⌜*then*⌝ would denote 'time concepts' modeled as partial functions from the index to a time interval:

(99) a. $[\![\textbf{PRES}_n]\!]^{c,g} = \lambda i : g(n) \circ i_t .\, g(n)$

b. $[\![\textbf{PAST}_n]\!]^{c,g} = \lambda i : g(n) < i_t .\, g(n)$

c. $[\![\text{then}_n]\!]^{c,g} = \lambda i : \neg(g(n) \circ i_t).\, g(n)$

First, note that identifying π with i_t has the same effect as making the shifting operator $\textbf{OP}_\pi$ *obligatory*. This immediately predicts that tense shift is obligatory in speech/attitude reports, as shown in (100). Since the embedded tense is now anchored to the index and the speech/attitude verb always binds the evaluation index of its complement, a shifted reading becomes unavoidable.

(100) Nate said Erica is angry. (Lit. from Modern Greek)

LF: $[\![$Nate say-$\textbf{PAST}_k$ [Erica be-$\textbf{PRES}_n$ angry]$]\!]^{c,g}$

a. $[\![$Erica be-$\textbf{PRES}_n$ angry$]\!]^{c,g} = \lambda i'.\, \textbf{ANGRY}(\text{g(n)})(\textbf{E})(i'_w)$ only if $g(n) \circ i'_t$

b. LF $\rightsquigarrow \lambda i.\, \forall i' \in \textbf{SAY}(g(k))(\textbf{N})(i_w).\, \textbf{ANGRY}(g(n))(\textbf{E})(i'_w)$

only if $\forall i' \in \textbf{SAY}(t)(x)(i_w).\, g(n) \circ i'_t (= g(k))$

This is an undesirable result. Even in languages where tenses *can* shift, the shifting is not obligatory. For example, certain temporal adverbials may force the present to be interpreted with respect to the time of utterance. This is shown in (101)—even though Modern Greek allows for a shifted present in reported speech under future, the embedded present can still be interpreted relative to the time of utterance if forced by the non-shiftable indexical *tora* 'now' (cf. (96)).[20]

(101) To 2030, i Maria **tha** pi oti **ine** sti filaki tora.
DET 2030, DET Maria **FUT** say.PFV that **be.PRES** to-the jail now
'In 2030, Maria will say that she is in jail now.' Modern Greek

Another reason not to identify the perspective with the temporal component of the index is that it would make the wrong prediction for deleted tense. Recall the possible LFs of deleted past with ⌜*then*⌝ in (90):

(90) a. $[\![$Mary miss-$\cancel{\textbf{PAST}}_2$ Nate then$_3]\!]^{c,\pi,g}$

$= \lambda i.\, \textbf{MISS}(i)(g(2))(\textbf{M})(\textbf{N}) \wedge \boxed{g(2) \subseteq g(3)}$, only if $\boxed{\neg(g(3) \circ \pi)}$

b. $[\![\textbf{OP}_\pi$ [Mary miss-$\cancel{\textbf{PAST}}_2$ Nate then$_3$]$]\!]^{c,\pi,g}$

$= \lambda i'.\, [\![$Mary miss-$\cancel{\textbf{PAST}}_2$ Nate then$_3]\!]^{c,i',g}$

$= \lambda i'.\, \textbf{MISS}(i')(g(2))(\textbf{M})(\textbf{N}) \wedge \boxed{g(2) \subseteq g(3)}$ only if $\boxed{\neg(g(3) \circ i'_t)}$

[20] An anonymous reviewer points out that the indexical reading of the present could be achieved in a perspective-as-index approach if we assume a distinguished variable $t*$ at the left periphery (von Stechow, 1995; Kusumoto, 2005) or a NOW operator (Kamp, 1967). Although this is a viable alternative that should be explored in future research, we prefer our system for reasons of theoretical simplicity. The $t*$/NOW operator would have to be constrained in some way, or else, if always there, it would often be redundant. Also, we do not think that adding a $t*$/NOW is necessarily simpler than adding a perspective, especially since the latter can explain a range of phenomena as will be seen in the next subsection (see also Sudo (2015) for locative perspectives for example).

Recall also that the simultaneous reading is obtained via the LF in (90a), where $\textsc{op}_\pi$ is absent. This is because when $\textsc{op}_\pi$ is present, the presupposition of *then*$_3$ is that it denotes a time distinct from the *shifted* perspective. Thus, when $\textsc{op}_\pi$ is present, *then*$_3$ cannot refer to a time simultaneous with the embedded verb. Instead, we need the LF in (90a) without the perspective shifting operator in order to get the simultaneous reading. But by identifying π with i_t and predicting obligatory tense shift, we lose the simultaneous reading. Indeed, if the perspective simply *is* the temporal component of the index, then both LFs would generate the same shifted reading as in (102), where *then*$_3$ has no way of denoting a time simultaneous with the time of the missing.

(102) a. $[\![\text{Mary miss-}\cancel{\textsc{past}}_2 \text{ Nate then}_3]\!]^{c,g}$
$= \lambda i.\, \textsc{miss}(i)(g(2))(\textsc{m})(\textsc{n}) \wedge \boxed{g(2) \subseteq g(3)}$, only if $\boxed{\neg(g(3) \circ i_t)}$

b. $[\![\textsc{op}_\pi\ [\text{Mary miss-}\cancel{\textsc{past}}_2 \text{ Nate then}_3]]\!]^{c,\pi,g}$
$= \lambda i'.\, [\![\text{Mary miss-}\cancel{\textsc{past}}_2 \text{ Nate then}_3]\!]^{c,g}$
$= \lambda i'.\, \textsc{miss}(i')(g(2))(\textsc{m})(\textsc{n}) \wedge \boxed{g(2) \subseteq g(3)}$, only if $\boxed{\neg(g(3) \circ i'_t)}$

Identifying the perspective with the index is thus unattainable, since it predicts obligatory tense shift, which is not the case, as well as fails to account for the deleted past. A main point of this paper is that the deleted past and the shifted present cannot be semantically equivalent, contrary to what is usually assumed. Identifying the perspective with the index collapses the shifted present and the deleted past, and thus fails to account for the incompatibility of ⌜*then*⌝ with the former but not with the latter.

In sum, reducing the perspective to either the temporal component of the context or the index will lead to undesirable predictions, and it is an important empirical claim of the paper that the interpretation of tenses calls for a *distinct* perspective parameter π.

7.3 More on the notion of perspective

Having established that we need a perspective parameter π, we may now ask whether π is always independent. It could be that π does not interact with other parameters at all, or that in certain contexts there are implicational hierarchies. Indeed, we already saw that in matrix clauses the perspective is co-valued with the temporal component of the context. In other words, by default, $\pi = c_t$. So, the perspective and the context parameters do interact. Are there other contexts where the perspective is forced to be co-valued with the temporal component of the context? We argue there are.

Kusliy (2017) explored the interaction between temporal indexicals, such as *today*, *yesterday* and *tomorrow*, and the shifted present. He observed that in Russian the shifted present can only co-occurs with the likewise shiftable temporal adverbial *segodnja* 'today', but not with the indexical *včera* 'yesterday', despite them both referring to the day before the time of utterance. This is illustrated by the examples below:

(103) *Context: Today is Tuesday. On Monday (i.e. one day earlier), Vanja utters: "Masha is sick today".*

a. Vanja skazal, čto Maša **segodnja bol'na**.
Vanja said that Masha **today be.PRES-sick**

"Vanja said that Masha was sick on that day."

b. #Vanja skazal, čto Maša **včera** **bol'na**.
Vanja said that Masha **yesterday be.PRES-sick**
Intended: "Vanja said that Masha was sick yesterday". Russian (Kusliy 2017: (20))

Similar effects are observed under future, an environment where as we saw tenses shift, since the future is a modal. In the English and Japanese examples below the embedded past is shifted and is only past relative to the 'now' of the attitude holder while being in the future with respect to the time of utterance. In such an environment where the tense is shifted, it is only compatible with a shiftable adverbial like *two days before*, but not with an indexical one like *tomorrow*.

(104) *Context: Today is Monday. Mary arrives on Tuesday. John talks about Mary's arrival on Thursday.*

a. John will say that Mary **arrived two days before**.

b. #John will say that Mary **arrived tomorrow**. (Kusliy 2017: (1))

(105) *Context: Today is Monday. Mary arrives on Tuesday. John talks about Mary's arrival on Thursday.*

a. Kazuko-wa [**hutu-ka-mae-ni** Tokyo-ni it-ta] otoko-ni a-w(-daroo).
Kazuko-TOP **two-day-before-in** Tokyo-to **go-PAST** man-DAT meet-PRES(-will)
'Kazuko will meet a man who went to Tokyo two days before.'

b. #Kazuko [**asu** Tokyo-ni it-ta] otoko-ni a-w(-daroo).
Kazuko-TOP **tomorrow** Tokyo-to **go-PAST** man-DAT meet-PRES(-will)
Japanese (Kusliy 2017: (2))

These data pattern well with the shift-together effect we also see with ⌜*then*⌝ and the present. When the indexical *včera* 'yesterday' is used in Russian, *asu* 'tomorrow' in Japanese, or *tomorrow* in English, the embedded tense is also interpreted indexically rendering the example infelicitous in the context. On the other hand, when the shiftable *segodnja* 'today' is used in Russian, *hutu-ka-mae-ni* 'two days before' in Japanese, or *two days before* in English, the embedded tense is also shifted. If the parameters are entirely independent in these environments, then we do not predict this shift together effect, since the tenses are evaluated with respect to the perspective and the temporal adverbials with respect to the context or the index parameters. Mismatches between the parameters are allowed, so we overgenerate readings for (103b), (104b) and (105b). To see this, take the example of (103b), repeated below as (106):

(106) #Vanja skazal, čto Maša **včera** **bol'na**.
Vanja said that Masha **yesterday be.PRES-sick**
Intended: "Vanja said that Masha was sick yesterday". Russian (Kusliy 2017: (20))

a. $[\![\text{yesterday}]\!]^{c,\pi,g}$ =the day before $\mathbf{c_t}$

b. $[\![\text{Vanja say-}\textbf{PAST}_1\ [\textbf{OP}_\pi\ \text{Masha be-}\textbf{PRES}_2\ \text{sick yesterday}]]\!]^{c,\pi,g}$

We currently allow for the LF in (106b), which contains the perspective-shifting operator $\textbf{OP}_\pi$ but not the context-shifting operator. Given the standard semantics in (106a) for *yesterday*, where it picks out the day before c_t, the embedded present $\textbf{PRES}_2$ can be shifted, while the temporal indexical is interpreted with respect to the context. To block this LF, we need to establish a dependency between perspective and context:

(107) When an overt temporal indexical occurs within a minimal clausal domain, the temporal perspective π is co-valued with the temporal component of the context c_t.

So it seems that further dependencies may exist between parameters. That being said, things could be more complicated than (107), since the there are indexicals in English can co-occur with a shifted tense, seemingly violating (107). Consider (108) from Kusliy (2017), where the indexical *today* is used with a shifted past.

(108) *Context: Charlie is expected to arrive today, although he hasn't at the time of utterance.*
Alice will tell Bob tomorrow that Charlie **arrived today**.

But in fact things are even more complicated. Not all temporal indexicals block a co-clausal shifted tense. For instance, Kusliy (2017) reported that (108), with the non-shifted indexical *today*, is felicitous in the given context. Since Charlie's arrival is expected *later* today, the embedded past can only be interpreted relative to a shifted perspective that presumably coincides with Alice's future speech. The fact that it is compatible with *today* clearly violates (107). Thus, (107) will need to be further refined, or else *today* could be shown to marginally have shiftable usages.[21] We leave a full exploration of the exact dependencies between parameters for later, but we hope to have opened an avenue on the interaction between (non-)shiftable adverbials and tense. Another interesting direction for future research is the dependency between the perspective and the index, as well as non-temporal usages of the perspective. For other perspective-sensitive expressions outside the temporal domain (Sudo, 2015), a more thorough treatment of the perspective parameter is needed, containing a speaker, an addressee, and a world, parallel to the context and the index.

Finally, other temporal adverbials can be analyzed as being perspective-sensitive as well, such as shiftable *now* in Russian. Vostrikova (2019) discusses interesting data where *sejčas* 'now' in Russian is incompatible with an embedded past interpreted *de re*, targeting a simultaneous reading:

(109) Kogda ja govorila s nej tri goda nazad, Tanja skazala, čto ona byla (***sejčas**) beremenna.
When I talk.PAST with her three years ago Tanja say.PAST that she be.**PAST** **now** pregnant
Intended: 'When I talked to her three years ago, Tanja told me that she was pregnant (at that time).'
Russian (Vostrikova 2019: (6))

[21] Kusliy argues that temporal indexicals as well as *then* 'force feed' their interpretation parameters (c_t for indexicals and i_t for *then*) to the co-clausal tense. He further distinguishes between what he calls (temporally) *oriented* adverbials (such as *yesterday*, *tomorrow*, *then*, *two days ago*) and *non-oriented* ones (such as *today*, *two days before*) claiming that *oriented* adverbials are (lexically) equipped to 'force feed' their parameter to the co-clausal tense. However, since he defines *oriented* adverbials using their incompatibility with a co-clausal tense, and uses this notion to then explain this incompatibility, the orientedness criterion becomes somewhat circular.

Unlike English *now*, *sejčas* 'now' in Russian can be shifted, referring to a time other than the time of utterance. This is shown in (110) below, where *sejčas* is compatible with a shifted present.[22]

(110) Kogda ja govorila s nej tri goda nazad, Tanja skazala, čto ona (**sejčas**) ∅ beremenna.
When I talk.PAST with her three years ago Tanja say.PAST that she **now** **be.PRES** pregnant
'When I talked to her three years ago, Tanja told me that she was pregnant (at that time).'
Russian (Vostrikova 2019: (5))

How can we look at these data under the perspective parameter lens? Recall the entry for the past from (71), repeated below as (111). Just like ⌜*then*⌝, *sejčas* 'now' can also be given a perspective-sensitive denotation:

(111) $[\![\textbf{PAST}_n]\!]^{c,\pi,g} = g(n)$ only if $g(n) < \pi$, otherwise undefined

(112) $[\![\text{sejčas}_n]\!]^{c,\pi,g} = g(n)$ only if $g(n) \circ \pi$, otherwise undefined

The incompatibility between the past and *sejčas* 'now' when they share the same perspective is then derived as a semantic anomaly, just like the ⌜*then*⌝-present puzzle. The simultaneous reading of the embedded past is derived with the following LF, where $\textbf{OP}_\pi$ does not apply and the reference of the embedded past coincides with that of the matrix past (i.e. $g(2) = g(1) < \pi$):

(113) $[\![$Tanja say-PAST$_1$ [she be.**PAST**$_2$ pregnant]$]\!]^{c,\pi,g}$

Finally, we would like to point out a difference between the perspective parameter and the context one, which should be explored in future research. Operator-based analyses of indexical shift make the nice prediction that a shiftable indexical cannot pick its reference from a context c if there is an intervening context c' from which another indexical picks its reference, as illustrated below:

(114) [Andrew:] Ali$_A$ mɨ$_U$-ra va kɛ Hɛseni$_H$ **to**$_U$**-ra** va ɛ**z**$_{H,A,{}^*U}$ braye Rojda-o
Ali me-to said that Hesen **you**-to said **I** brother Rojda-GEN
'Ali said to Andrew that Hesen said to Andrew that {Hesen, Ali, *Andrew} is Rojda's brother.'
Zazaki (Anand and Nevins 2004: (32))

[22] There is within-speaker variation with respect to whether Russian speakers have a shiftable 'now'. We tested (1) with two consultants, one of which had an indexical 'now', while the other could access a shifted reading:

(1) V 2000m godu Ivan znal, čto Maša (%sejčas) beremenna.
In 2000 year Ivan know.PRES that Masha now pregnant
'In 2000, Ivan knew that Masha was (LIT.: is) pregnant.'

More specifically, the consultant with the indexical reading found (1) odd, and the only possible reading they could access involved clairvoyance (like in Greek, Japanese). The other consultant initially had an indexical reading of 'now', but after some thought could also access a shifted reading of both 'now' and the embedded present, as in the following (p.c. Katya Murgonova):

(2) Ivan znal, čto sejčas Maša beremena i nuzhno ekonomit' den'gi
Ivan **know.PAST** that now Masha pregnant and need-to save money
'Ivan knew that Masha was (LIT.: is) pregnant then (LIT.: now) and that they should save up money.'

The consultant also mentioned that under the shifted reading, 'now' cannot be prosodically focused. We treat some Russian speakers as having the entry of the English/Greek indexical 'now', and some as having the shiftable *sejčas* 'now' Vostrikova describes.

In this example there are two shifted indexicals: the embedded second-person indexical *to* 'you' that is shifted to the addressee in the reported speech (who happens to be the actual speaker Andrew), and a more deeply embedded first-person indexical *ϵz* 'I'. Crucially, with the intermediate indexical *to* 'you' *shifted*, the lowest indexical *ϵz* 'I' can no longer draw reference from the matrix utterance context; instead, it has to be interpreted relative to the intermediate speech context (whose speaker is Ali), or to be further shifted, referring to the speaker of the lowest reported speech, namely Hesen. This is because once the operator overwrites the context parameter, the information about the matrix context is lost. This is known as the No-INTERVENING-BINDER constraint. As a reviewer points out, we make the same prediction in the temporal domain, since once we shift the perspective we can no longer retrieve the original perspective parameter. To test this prediction consider the following examples containing two presents under future:[23]

(115) John will announce that he **is** working with someone who **is** working with the government.

(116) O Yanis tha anakinosi oti dhulevi me kapion pu dhulevi stin kivernisi.
DET Yanis FUT announce.PFV COMP work.PRES.IMPFV.3SG with someone.ACC who work.PRES.IMPFV.3SG to-the government

a. ✓Yanis: "I am working with someone who **is** working for the government" (both shifted)

b. ?? Yanis: "I am working with someone who **was** working for the government" (shifted, indexical)

c. ✓Yanis: "I **was** working with someone who **was** working for the government" (both indexical)

d. ✗Yanis: "I **was** working with someone who **is** working for the government" (indexical, shifted)

The current operator-based analysis predicts that both presents should be shifted or both should be indexical; it does not allow for mixing of perspectives. This correctly rules out (116d), while also predicting ungrammaticality for (116b). In (116b) the first present is shifted, while the second is read indexically; this reading is only available in a scenario where the second present is true at the time of the utterance *and* at the time of the speech as in (117a) and (117b) below. It is impossible if the second present is only indexical and is not true at the time of the speech, as shown in (117c) and (117d). Thus, we believe that a shift together effect is indeed at play here.

(117) O Yanis tha anakinosi oti dhulevi me kapion pu dhulevi stin kivernisi.
DET Yanis FUT announce.PFV COMP work.PRES.IMPFV.3SG with someone.ACC who work.PRES.IMPFV.3SG to-the government

a. First present indexical and shifted, Second present indexical and shifted ✓

Yanis has been working for many months with the minister of agriculture. Yanis will announce it at the company's meeting next Fall.

[23] We thank Amy Rose Deal for helpful discussions on this and similar issues during WCCFL, as well as an anonymous reviewer who also points this out.

b. First present only shifted, Second present indexical and shifted ✓
Yanis will start working with the minister of agriculture in 2 months. He will announce it at the company's meeting next Fall.

c. First present only shifted, Second present only indexical ✗
The current minister of agriculture plans to resign from his government position and start working with Yanis. Yanis and the current minister have signed a contract, and will start working together in 2 months. Yanis plans to announce it at the company's meeting next Fall.

d. First present indexical and shifted, Second only indexical ✗
The current minister of agriculture is working with Yanis, whose company is a government contractor. The minister plans to resign from his government position and continue working exclusively with Yanis. Yanis plans to announce it at the company's meeting next Fall.

We have delved deeper into the perspective parameter and discussed ways in which it can be further explored, in the temporal domain and beyond. In Appendix B, we discuss another potential extension of the perspective parameter to the modal domain.

8 Conclusion

This paper delved into the observation that ⌜*then*⌝ is incompatible with a shifted present, but compatible with deleted tense. To account for this puzzle, we proposed a new theory of tense and tense shift, centered around the notion of a temporal *perspective*. The temporal perspective is an interpretation parameter to which the interpretations of tenses are anchored; it can be shifted by a propositional perspective-shifting operator resulting in tense shift. We have shown that this new theory provides an elegant solution to the ⌜*then*⌝-present puzzle, while also capturing the difference between shifted present and deleted tense. Although we only scratched the surface of the nature of the temporal perspective, we emphasized that it should be distinguished from more familiar parameters such as the context and the evaluation index. Formally, our system mirrors the operator-based analysis of indexical shift (Anand and Nevins 2004; Anand 2006), predicting a shift together effect for tenses. We hope that future research will incorporate tense to the growing body of literature establishing implicational hierarchies of shiftable, context- and perspective-sensitive expressions cross-linguistically (e.g., Deal 2017, 2020, Sundaresan 2018, 2021).

Statements and Declarations

The authors have no competing interests to declare that are relevant to the content of this article.

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Appendix A: Shifted present in Japanese relative clauses

Japanese is the only language examined in this paper that allows the present to shift in an extensional environment, namely in relative clauses under past. Usually, tense shift occurs in intensional environments, either under a speech/attitude verb or under future. Japanese is the only exception. How exactly the Japanese present can shift in extensional environments is an independent puzzle from the ⌜*then*⌝-present puzzle, which we do not attempt to solve here. However, the ⌜*then*⌝-present puzzle could be a useful clue towards its solution. More specifically, the ⌜*then*⌝-present puzzle persists in Japanese relative clauses. This in turn suggests, that whatever is responsible for the puzzle in intensional environments in all these other languages (including in Japanese attitude reports) is also responsible for the puzzle here. In other words, the persistence of the puzzle suggests that the same tense shifting mechanism is at work in Japanese relative clauses. We will therefore propose a similar explanation for the ⌜*then*⌝-present puzzle in Japanese relative clauses, provided that the present has an (independent from ⌜*then*⌝) way to shift in these environments.

Let us start with the following pseudo-glossing as a working example:

(118) Nate met someone who is in jail (*then). (Lit. from Japanese)
Intended: Nate met someone who was in jail (at the time they met).

Like before, the embedded present shifts thanks to $\textbf{OP}_\pi$, which composes with the relative clause. The difficulty here is that the main verb *meet* is extensional, and therefore cannot bind the index. How can the higher tense bind the local perspective at the absence of an attitude verb? For the purposes of this paper, we propose that tenses in Japanese are themselves intensional and can thus bind the temporal index.[24] More specifically, we assume that Japanese tenses (i) specify the time of the event/state, as tenses usually do, and (ii) bind the time component i_t of the index i used to evaluate their prejacent. It is this latter function that separates Japanese tenses from other tenses we have seen so far. In (119) and (120) below, we provide a lexical entry for the Japanese present and past. They are characterized as partial functions that take as arguments a

[24]One does not need to agree with this solution to accept our theory of tense shift. Whatever other solution works to shift the Japanese present in an extensional environment can also be implemented in our system.

time interval t (which has to satisfy a perspectival presupposition), an intensional temporal property P (their untensed prejacent), and an evaluation index i, and return true iff P holds for t when evaluated at the index $i[i_t \mapsto t]$, which is identical to i except that the time component i_t is updated to t.[25]

(119) $[\![\textbf{PRES}_{\textbf{JAP}}]\!]^{c,\pi,g} : \mathtt{i}{\rightarrow}(\mathtt{k}{\rightarrow}\mathtt{i}{\rightarrow}\mathtt{t}){\rightarrow}\mathtt{k}{\rightarrow}\mathtt{t}$

$$:= \lambda t_{\mathtt{i}} : t \circ \pi.\, \lambda P_{\mathtt{k}\rightarrow\mathtt{i}\rightarrow\mathtt{t}}\,.\lambda i_{\mathtt{k}}.\, P(i[i_t \mapsto t])(t)$$

where $i[i_t \mapsto t]$ an evaluation index just like i, except that i_t is substituted with t (same below).

(120) $[\![\textbf{PAST}_{\textbf{JAP}}]\!]^{c,\pi,g} : \mathtt{i}{\rightarrow}(\mathtt{k}{\rightarrow}\mathtt{i}{\rightarrow}\mathtt{t}){\rightarrow}\mathtt{k}{\rightarrow}\mathtt{t}$

$$:= \lambda t_{\mathtt{i}} : t < \pi.\, \lambda P_{\mathtt{k}\rightarrow\mathtt{i}\rightarrow\mathtt{t}}\,.\lambda i_{\mathtt{k}}.\, P(i[i_t \mapsto t])(t)$$

The LF of the working example (118) is given in (121). It makes explicit that (i) the matrix past tense takes scope over the untensed component of the root clause (similarly, the embedded present tense takes scope over the untensed component of the relative clause), (ii) the generalized quantifier in the object position containing the relative clause is QR'd, but still under the scope of the matrix past, and (iii) $\textbf{OP}_\pi$ is applied at the propositional node of the relative clause.

(121) *Nate met someone who is in jail then.

$[\![\textbf{PAST}_{\textbf{JAP}}\text{-}t_0$

$\quad\lambda_1$ [some (person)

$\quad\quad\lambda_2$ [$\textbf{OP}_\pi$[$\textbf{PRES}_{\textbf{JAP}}$-$t_3$ [λ_4[x_2 be-t_4 in jail then$_5$]]]]

$\quad\quad\quad\lambda_6$ [Nate meet-t_1 x_6]]]$]\!]^{c,\pi,g}$

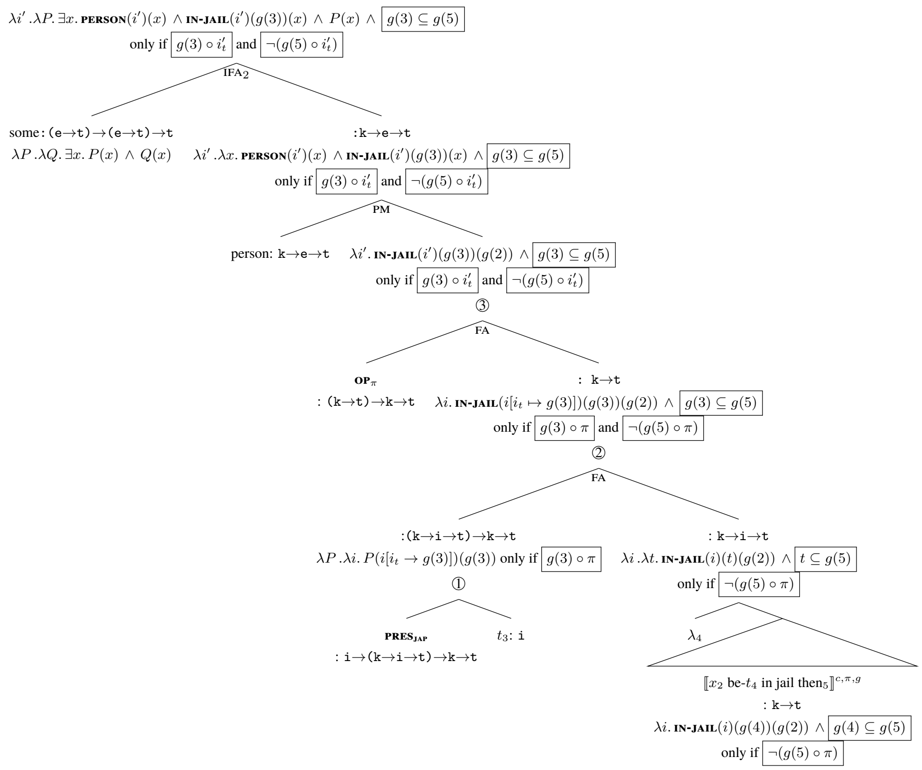
The derivation of the generalized quantifier restricted by the relative clause is in (122). At step ① $\textbf{PRES}_{\textbf{JAP}}$ takes the time argument t_3, which overlaps with the (local) perspective, per the entry in (119). Then at step ②, the denotation of t_3, namely $g(3)$, is fed to the predicate *be in jail*, while also used to substitute the temporal component of the index i_t. Thus, the relative clause ends up being evaluated at $g(3)$. At step ③, $\textbf{OP}_\pi$ applies and overwrites the perspective parameter with an abstracted index i', as usual. This leads to the shifting of ⌜*then*⌝ and the present. Recall that this is an unattested LF of an ungrammatical sentence. This is why a semantic anomaly is predicted, the same one we have been seeing with all combinations of ⌜*then*⌝ with the present. More specifically, ⌜*then*⌝ and the present share the same temporal perspective i'_t, and a contradiction arises between presupposed and asserted content.[26] The rest of the derivation is standard, since the quantifier has the usual type $(\mathtt{e}{\rightarrow}\mathtt{t}){\rightarrow}(\mathtt{e}{\rightarrow}\mathtt{t}){\rightarrow}\mathtt{t}$, and the relative pronoun *who* is treated as a lambda-abstractor λ_2.

[25] This analysis follows in spirit many existing theories that attempt to account for tense shift in Japanese. For instance, Ogihara (1996, 2022) argued that Japanese tenses are purely relative (e.g. should be taken as tripartite relations of two time arguments and a temporal property), and Ogihara and Sharvit (2012); Sharvit (2014) suggested that Japanese tenses are quantificational, as opposed to English pronominal tenses, while tense shift is achieved through the higher tense taking scope over the shifted one. The only reason we do not directly use one of these existing accounts is that they often adopt (explicitly or implicitly) a framework that treats temporal anchors as pronouns. As we argued for in Section 2.3, a parametric (as opposed to a pronominal) account of the temporal anchor is best suited to account for the ⌜*then*⌝-present puzzle.

[26] Note that since $\textbf{OP}_\pi$ re-abstracts over the evaluation index that the embedded present had already abstracted over, the effect of the first abstraction is essentially lost.

(122) someone who is$_3$ in jail then$_5$

$[\![$some (person) λ_2 [**OP**$_\pi$[**PRES$_{\text{JAP}}$**-t_3 [λ_4[x_2 be-t_4 in jail then$_5$]]]]$]\!]^{c,\pi,g}$: k→(e→t)→t



The final steps of the derivation are in (123). The quantifier takes scope over the untensed matrix clause (step ④), and the result combines with the matrix past in step ⑤. The matrix past tense is evaluated with respect to the time of utterance $g(0)$, situating the meeting event prior to $g(0)$. Finally, because we assume that Japanese tenses can bind indices, the past binds the time component of the index, i'_t, replacing it with $g(0)$. Thus, ⌜*then*⌝ and the embedded present end up being both anchored to $g(0)$. Since they have the same anchor, and conflicting presuppositions, the same semantic anomaly as before arises.

(123) $[\![{}^{*}\text{Nate met}_0 \text{ someone who is}_3 \text{ in jail then}_5]\!]^{c,\pi,g} \texttt{:k}\!\rightarrow\!\texttt{t}$

$\lambda i.\, \exists x.\, \textsc{person}(i)(x) \wedge \textsc{in-jail}(i)(g(3))(x) \wedge \textsc{meet}(i)(g(0))(x)(\textsc{n}) \wedge \boxed{g(3) \subseteq g(5)}$
only if $\boxed{g(3) \circ g(0)}$ and $\boxed{\neg(g(5) \circ g(0))}$

⑤
FA

$\texttt{:(k}\!\rightarrow\!\texttt{i}\!\rightarrow\!\texttt{t)}\!\rightarrow\!\texttt{k}\!\rightarrow\!\texttt{t}$
$\lambda P\,.\lambda i.\, P(i[i_t \rightarrow g(0)])(g(0))$ only if $\boxed{g(0) < \pi}$
FA
$\textsc{past}_{\textsc{jap}}\ \texttt{:i}\!\rightarrow\!\texttt{(k}\!\rightarrow\!\texttt{i}\!\rightarrow\!\texttt{t)}\!\rightarrow\!\texttt{k}\!\rightarrow\!\texttt{t}$ $\quad t_0\ \texttt{:i}$

$\texttt{:k}\!\rightarrow\!\texttt{i}\!\rightarrow\!\texttt{t}$
PA
λ_1

Nate meet-t_1 someone who is$_3$ in jail then$_5$ $\texttt{:k}\!\rightarrow\!\texttt{t}$
$\lambda i'.\, \exists x.\, \textsc{person}(i')(x) \wedge \textsc{in-jail}(i')(g(3))(x) \wedge \textsc{meet}(i')(g(1))(x)(\textsc{n}) \wedge \boxed{g(3) \subseteq g(5)}$
only if $\boxed{g(3) \circ i'_t}$ and $\boxed{\neg(g(5) \circ i'_t)}$

④
IFA$_3$

someone who is$_3$ in jail then$_5$
(122)

$\texttt{:k}\!\rightarrow\!\texttt{e}\!\rightarrow\!\texttt{t}$
$\lambda i\,.\lambda x.\, \textsc{meet}(i)(g(1))(x)(\textsc{n})$
PA
$\lambda_{6:\texttt{e}}$

Nate meet-t_1 x_6 $\texttt{:k}\!\rightarrow\!\texttt{t}$
$\lambda i.\, \textsc{meet}(i)(g(1))(g(6))(\textsc{n})$

Making Japanese tenses able of binding indices is a technical solution, compatible with our system, to shift the present in an otherwise extensional environment. However, as already mentioned, we are not wed to this solution. The point that we want to make in this paper is that ⌜*then*⌝ and the present have conflicting presuppositions, while ⌜*then*⌝ and deleted past do not. And for the Japanese present, what this amounts to saying is that a shifted present will always be compatible with ⌜*then*⌝ no matter how it shifted.

Appendix B: Further Extensions

We show that the ⌜*then*⌝-present puzzle extends straightforwardly to a present tense embedded under a modal. Then, we discuss a case of a morphological present in Modern Greek, which is semantically non-past, and thus is predicted to be compatible with ⌜*then*⌝.

Extensions to present embedded under modals

We would like to bring up an interesting extension to the puzzle in present embedded under modals. As Condoravdi (2002) shows, modals can be future-oriented. For example, the following sentence in English is ambiguous between a present- and a future-oriented interpretation:

(124) He may be here.

a. ✓He may be here (right now).

b. ✓He may be here (tomorrow).

Interestingly, when we add ⌜*then*⌝ this ambiguity disappears, with only the future oriented reading being accessible:[27]

(125) He may be here then.

a. ✗He may be here (right now).

b. ✓He may be here (tomorrow).

In the sentence above, the modal can only refer to a future time. The same pattern holds for past tense modals, which without ⌜*then*⌝ are three-way ambiguous between (i) a past tense reading, (ii) a counterfactual epistemic present tense one, and (iii) a future tense reading.

(126) He could have been here.

a. ✓He could have been here (yesterday)

b. ✓He could have been here (now).

c. ✓He could have been here (tomorrow).

Again, the present tense interpretation is excluded the moment we add ⌜*then*⌝:

(127) He could have been here then.

a. ✓He could have been here (yesterday)

b. ✗He could have been here (now).

c. ✓He could have been here (tomorrow).

Thus, it seems that modals that are ambiguous between a present tense interpretation and some other interpretation (past or future) lose the former once ⌜*then*⌝ is added. Thus, the ⌜*then*⌝-present puzzle seems to hold with modals too. The same holds for a present tense inside an if-clause, which is ambiguous between a present- and a future-oriented reading:

(128) If he is here, …

a. ✓If he is here now, …

b. ✓If he is here tomorrow, …

[27] We thank Cleo Condoravdi for pointing this out to us during WCCFL.

Again, once ⌜*then*⌝ is added we lose the present tense reading:

(129) If he is here then, …

  a. ✗If he is here now, …

  b. ✓If he is here tomorrow, …

We thus see that the ⌜*then*⌝-present puzzle also extends to a present under a modal, as we would predict, since our theory predicts incompatibility between any (semantically contentful) present tense and ⌜*then*⌝.

Non-past vs. present

In our theory, we account for the ⌜*then*⌝-present puzzle thanks to the perspectival presuppositions of the present and ⌜*then*⌝, which when evaluated with the same perspective yield a contradiction. Crucially, we do not predict *any* morphological present to be incompatible with ⌜*then*⌝, only those that semantically presuppose an overlap with the perspective.[28] For example, we already saw that the English present embedded under *will* is a deleted present, and is thus compatible with *then*. A similar phenomenon also exists in Modern Greek in the form of a semantically nonpast, morphologically present tense. In our system, the non-past is correctly predicted to be compatible with ⌜*then*⌝. Subjunctive mood in Modern Greek is marked with the *na* particle, such as in the following sentence:

(130)
I	Maria	theli	na	jrapsi	ena	vivlio.
DET	Maria	want.PRES	SUBJV	write.PERF.NONPAST	a	book

‘Maria wants to write a book.’

Giannakidou (2009) noted that the tense on the embedded verb is a perfective nonpast, even though morphologically it surfaces as a present tense. She characterizes this as “a dependent verbal form with no formal mood features: the perfective nonpast (PNP)” (see also Holton et al. (2012)). PNP is a dependent form, since it cannot appear in root clauses (except under a future modal), and it can only surface inside a subjunctive clause. There is also no imperfective equivalent of the nonpast.

As Giannakidou (2009) notes, the PNP should semantically be distinguished from the present (even though morphologically PNP surfaces as a present). For our purposes, this means that PNP has present tense morphology without present tense semantics, i.e., without presupposing an overlap with the temporal perspective. Thus, we would predict it to be compatible with ⌜*then*⌝. Indeed, this prediction is borne out:[29]

(131)
To	2000,	o	Yanis	ithele	i	Maria	na	mini	egkios	tote.
DET	2000	DET	Yanis	want.PAST	DET	Maria	SUBJV	get.PNP	pregnant	then

‘In 2000, Yanis wanted Maria to get pregnant then.’

We thus see that the ⌜*then*⌝-present puzzle is a puzzle about a semantically (and not simply a morphologically) present tense. Our theory correctly predicts this.

[28] We also do not *require* present tense morphology for the puzzle to arise. If another morphological tense is a present in disguise having the same presupposition, we also predict incompatibility with ⌜*then*⌝.

[29] We thank Anastasia Giannakidou for pointing this out to us during CLS.